

# Estimates Difference of Lunar Regolith Thermal Conductivity $K_d$ at the Apollo 15 and 17 Heat-Flow Boreholes

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## Abstract

The Apollo 15 and 17 landing sites are the only locations on the Moon where subsurface regolith temperatures have been measured in situ. The restored 1971–1977 Heat-Flow Experiment (HFE) record provides the sole direct constraint on the meter-scale regolith thermal conductivity,  $K_d$ , yet current thermal models apply a single Moon-wide value, for example in the calibration of orbital Diviner brightness-temperature retrievals (Hayne et al., 2017; Martínez and Siegler, 2021). We retrieve  $K_d$  separately at each borehole from the restored record (Nagihara et al., 2018) by sweeping  $K_d$  against the meter-scale-sensor RMSE (80–139 cm at Apollo 15; 130–243 cm at Apollo 17) under the Hayne et al. (2017) form, using automatically selected per-sensor stability windows and excluding borestem-contaminated shallow sensors *a priori*. Conditional on the published basal heat fluxes (21 and 15 mW m<sup>-2</sup>; Langseth et al., 1976), the retrieval yields  $K_{d,A15}^* = 4.58^{+1.33}_{-0.28}$  and  $K_{d,A17}^* = 8.12^{+0.49}_{-0.61}$  mW m<sup>-1</sup> K<sup>-1</sup> (1 $\sigma$  non-parametric bootstrap), reducing the meter-scale-sensor RMSE from 1.03 and 0.69 K under the global value  $K_d = 3.4$  to 0.95 and 0.34 K. The inter-site contrast,  $\Delta K_d^* = 3.3^{+0.7}_{-1.3}$  mW m<sup>-1</sup> K<sup>-1</sup> (95% CI [0.4, 4.6]), excludes zero at the published basal heat fluxes ( $p \approx 0.01$ ), although propagating the basal-flux reanalysis envelope dilutes the ordering probability to  $\sim 83\%$ . Both retrieved values lie within porosity, composition, and orbital benchmarks. The per-site  $K_d^*$  values supply the meter-scale temperature-profile boundary condition required by future subsurface radiative-transfer retrievals, including the TSUKIMI terahertz mission, and an in-situ reference for future global conductivity models.

**Plain Language Summary.** The Moon’s interior is insulated from the daily surface heating by the compacted regolith (broken rock and dust) in roughly the top meter. The numbers describing that insulation come from formulas fit to satellite measurements of the lunar surface temperature, not to any direct sub-surface data. Sub-surface temperatures were measured at only two locations, by thermometers emplaced ~~more than two meters deep~~ up to 1.4 m (Apollo 15) and 2.4 m (Apollo 17) deep by the Apollo 15 and 17 astronauts in 1971–1972, and the complete records were recently recovered from the original mission tapes. We use those two records to determine, separately at each borehole, the regolith conductivity ~~at depths below about 30 cm~~ in the meter-scale subsurface (80–139 cm for Apollo 15 and 130–243 cm for Apollo 17), where the daily surface temperature swing has died away and the conductivity has settled to a steady value. The Hayne (2017) global value misses the ~~deep-meter-scale equilibrium~~ temperatures by about ~~1.4 K at Apollo 15 and 3.0 K at Apollo 17~~; ~~site-by-site refitting reduces those errors to  $\lesssim 1$  K~~ 1.0 K at Apollo 15 and 0.7 K at Apollo 17; ~~site-by-site refitting reduces those errors, halving the Apollo 17 mismatch~~, and the two sites disagree on the ~~deep-meter-scale~~ conductivity by roughly a factor of ~~two~~ 1.8. The site-fit values provide the ~~deep-temperature meter-scale temperature~~ input that upcoming lunar sub-surface mapping missions will need to interpret what their instruments observe through the regolith.

## Key Points

- ~~Deep  $K_d$  (80–234 cm) retrieved per-site at Apollo 15 and 17 after per-sensor stability-window selection of the restored 1971–1977 HFE record.~~ Meter-scale regolith conductivity  $K_d$  is retrieved per site from the restored Apollo 15 and 17 heat-flow records (80–234 cm).
- ~~The Apollo 15 and 17 HFE retrieval gives regolith  $K_d^{\Delta 15} = 4.86^{+1.12}_{-0.54}$  and  $K_d^{\Delta 17} = 11.23^{+8.12}_{-0.77}$  mW m<sup>-1</sup> K<sup>-1</sup> (1 $\sigma$  bootstrap).~~ Apollo 15 gives  $K_d = 4.58^{+1.33}_{-0.28}$  and Apollo 17 gives  $8.12^{+0.49}_{-0.61}$  mW m<sup>-1</sup> K<sup>-1</sup> (1 $\sigma$  bootstrap).
- ~~A single Moon-wide  $K_d$  is statistically rejected; future lunar thermal models and missions require site-specific values.~~ At published basal heat fluxes a single Moon-wide  $K_d$  is rejected ( $p \approx 0.01$ ); meter-scale models need site-specific values.

## 1 Introduction

Lunar sub-surface thermal modelling presently relies on globally averaged conductivity parameterizations. We use the term *deep-meter-scale regolith* throughout to mean ~~depths  $z \gtrsim 30$  cm, where the daily surface heat wave has damped to undetectable amplitude~~ the subsurface layer constrained by the Apollo HFE temperature sensors used in this retrieval—specifically, 80–139 cm for Apollo 15 and 130–243 cm for Apollo 17. At these depths the daily and annual thermal skins ( $\lesssim 10$  cm and  $\sim 50$  cm respectively) have damped to negligible amplitude, and the conductivity in both standard parameterizations has ~~effectively reached~~ converged on its depth-independent asymptote  $K_d$ ; ~~the daily skin itself spans  $\lesssim 10$  cm. The deep-regolith under the Hayne (2017) form, the same  $K_d$  value extrapolates to all greater depths within the regolith column.~~ The meter-scale regolith conductivity is supplied by formulations calibrated against orbital Diviner brightness temperatures: the smooth-exponential  $K(T, z)$  of Hayne et al. (2017), in which a single ~~deep-regolith-meter-scale~~ asymptote  $K_d$  is fit to a global average and applied moon-wide, and the temperature- and density-dependent  $K(T, \rho)$  of Martínez and Siegler (2021), which extends the same functional family with additional physical content but is itself calibrated globally and applied uniformly. Both parameterizations probe only the diurnal skin. The extent to which the resulting global  $K_d$  represents any individual landing site, and the magnitude of any site-to-site departure, cannot be determined from orbital data alone.

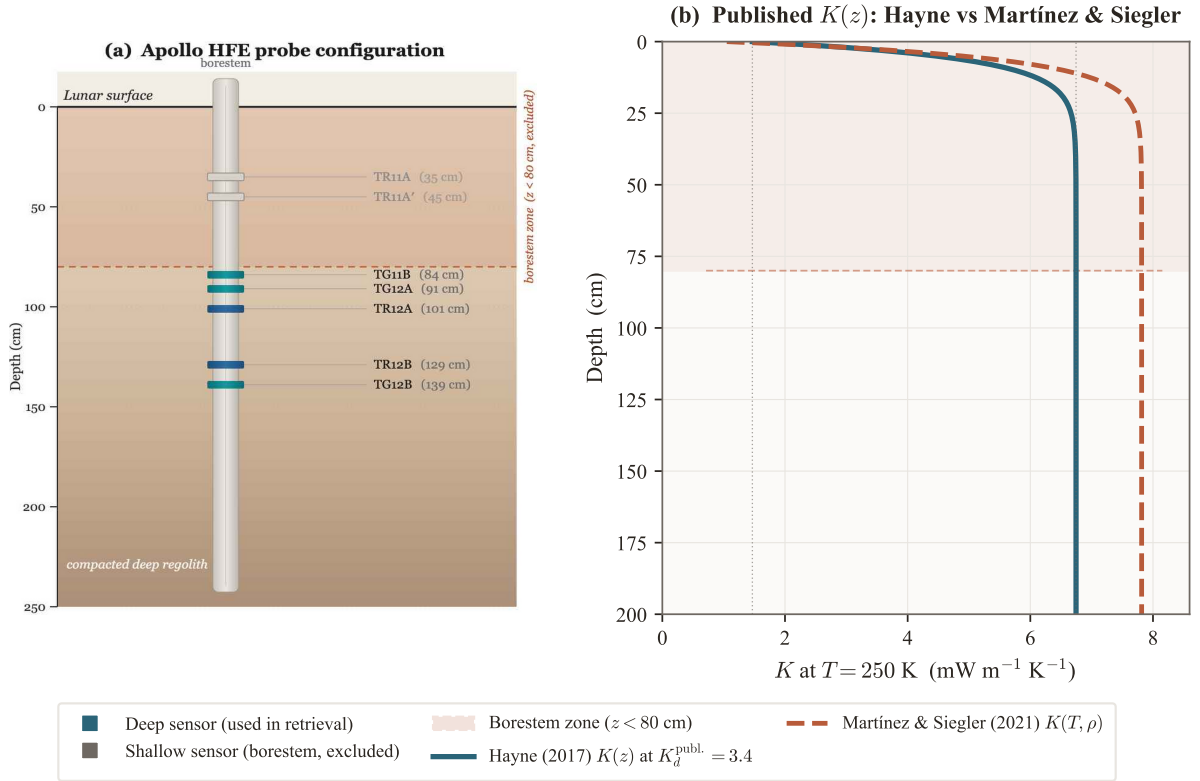
Direct in-situ data exist only from the Apollo Heat-Flow Experiment (HFE). At Apollo 15 and 17, fibreglass borestems emplaced by the astronauts carried thermometers to depths exceeding two meters (Langseth et al., 1972, 1976); these two boreholes remain the only lunar locations at which a sub-surface temperature profile has been measured (Fig. 1). The restored 1971–1977 HFE record (Nagihara et al., 2018) provides the data used here. Each borehole also carries a published basal heat flux (Langseth et al., 1976; Nagihara et al., 2018) which, together with the local  $K_d$ , sets the ~~deep~~ meter-scale temperature gradient. The site-specific inputs (latitudes, albedos, emissivities, basal fluxes) are collected with their original sources in Table 1.

We retrieve  $K_d$  separately at each site by holding the Hayne et al. (2017) functional form fixed and sweeping  $K_d$  alone against the ~~deep-sensor-meter-scale-sensor~~ RMSE. The retrieval is placed in context by two additional forward evaluations against the same ~~deep-sensor-meter-scale-sensor~~ set: the Hayne et al. (2017) form at its published global  $K_d$ , and the Martínez and Siegler (2021)  $K(T, \rho)$  model at its published coefficients (Table 2). Neither published model is refit here, so the Martínez and Siegler (2021) comparison quantifies the performance of the published global  $K(T, \rho)$  at each Apollo site without per-site freedom. A full per-site retrieval under the Martínez and Siegler (2021) form requires a different choice of free parameter from  $K_d$  and is deferred to a follow-on study (§4.3).

The per-site  $K_d^*$  values matter for any sub-surface radiative-transfer (RTM) retrieval that observes the Moon at wavelengths penetrating below the diurnal skin: future microwave, ~~terahertz, and infrared~~ and ~~terahertz~~ sub-surface sounders all need a ~~deep-meter-scale~~ temperature column  $T(z)$  as a forward-model boundary condition that orbital instruments cannot measure directly. One concrete example is the upcoming TSUKIMI mission (Lunar Terahertz Surveyor for Kilometer-scale Mapping, Kasai et al., 2022), whose ~~terahertz radiometer will probe depths comparable to the Apollo HFE sensors and so depends explicitly on  $K_d$~~  forward model requires the sub-diurnal-skin temperature column that only the in-situ  $K_d$  can anchor.

Because the emission-weighting kernel of any such instrument reaches depths at which the conductivity has approached its asymptote  $K_d$ , the retrieved  $T(z)$  depends on  $K_d$  and not only on the shallow-skin  $K_s$ ; aBelow the diurnal skin the modelled column is controlled by  $K_d$  through both the rectified annual mean and the steady gradient  $Q_b/K_d$ , so the  $T(z)$  ingested by any such retrieval inherits its  $K_d$  dependence; a factor-of-two error in  $K_d$  propagates through the deep-meter-scale temperature profile and into any retrieval that uses it. The values reported here, evaluated in the Hayne et al. (2017) thermal column, supply that boundary condition from the only in-situ data available.

The retrieval is driven by an ephemeris-accurate solar-forcing chain (JPL DE440 lunar and solar positions accessed through SPICE; Acton 1996, 2018), which eliminates the  $\sim 1$  lunar-hour drift inherent in sinusoidal local-solar-time approximations and is required by the phase-folded diurnal-amplitude profile underlying the borestem cut (§3.1). The retrieval is driven by an idealized synodic insolation cycle evaluated at each site’s latitude (§2); because only stability-window mean temperatures enter the fit, sub-cycle timing plays no role, and the neglected ephemeris terms (solar declination, orbital eccentricity) perturb the cycle-mean insolation at or below the percent level (§2). The remaining sections present the data and methods (§2), the per-site retrieval with the headline RMSE table (§3), the interpretation with its uncertainty limits (§4), and the conclusions.

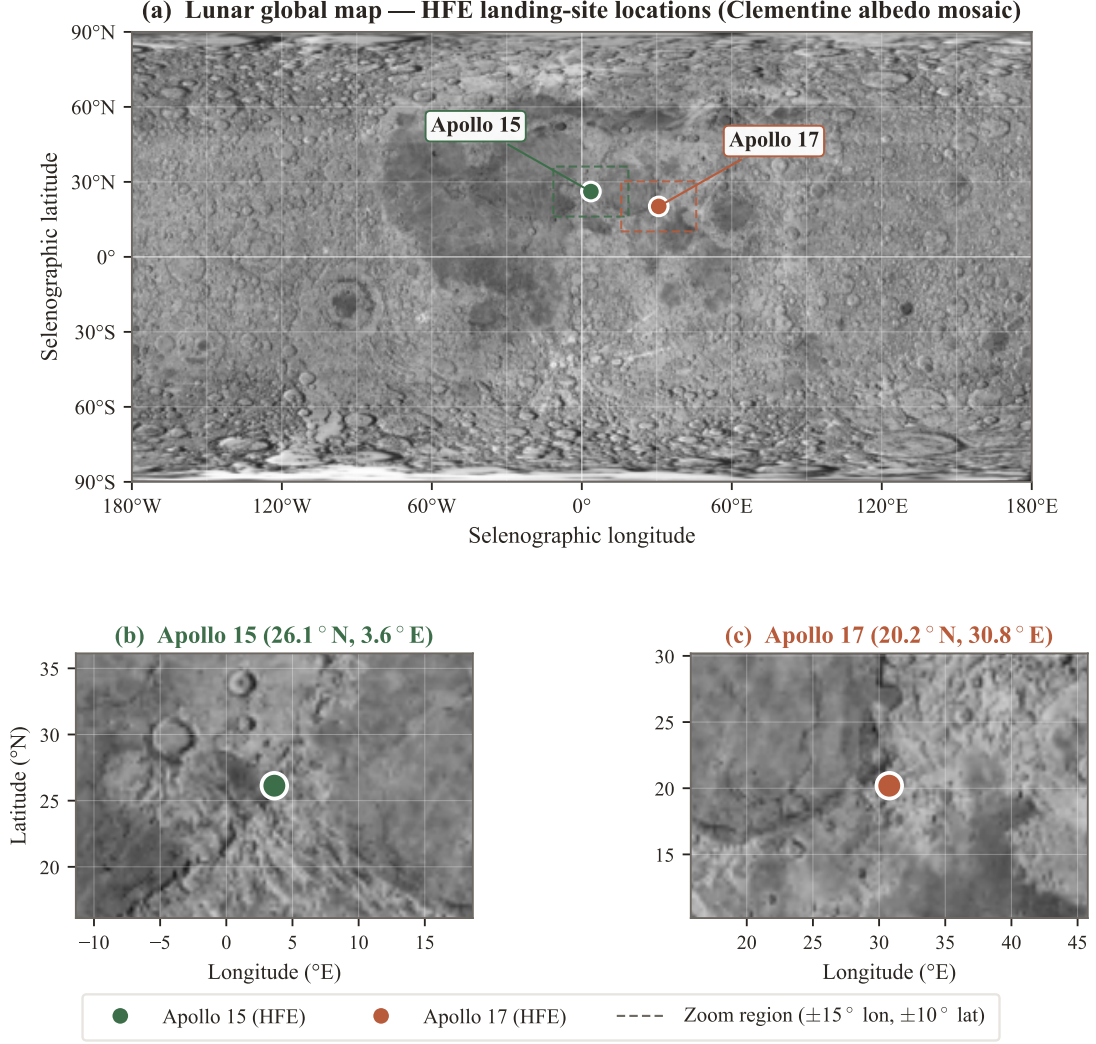


**Figure 1.** Apollo HFE measurement geometry and the two published  $K(z)$  forms tested. **(a)** Probe schematic showing the fibreglass borestem and thermometer placements (Apollo 15 Probe-1). Coral band: borestem-excluded zone ( $z < 80$  cm); teal markers: deep-meter-scale sensors used in the retrieval. **(b)** Thermal conductivity profiles  $K(z)$  at  $T = 250$  K: Hayne et al. (2017) smooth-exponential form at  $K_d^{\text{publ.}} = 3.4 \text{ mW m}^{-1} \text{K}^{-1}$  (solid teal); Martínez and Siegler (2021)  $K(T, \rho)$  at published coefficients (dashed red). Both use the same Hayne-form  $\rho(z)$ .

## 2 Data and Methods

### 2.1 Apollo HFE meter-scale-sensor temperatures

The retrieval requires one equilibrium temperature  $T_{\text{eq}}$  per sensor. The restored 1971–1977 record (Nagihara et al., 2018) provides this quantity once the early portion of each record, which contains residual transients from drilling and emplacement, documented mission-operations disturbances



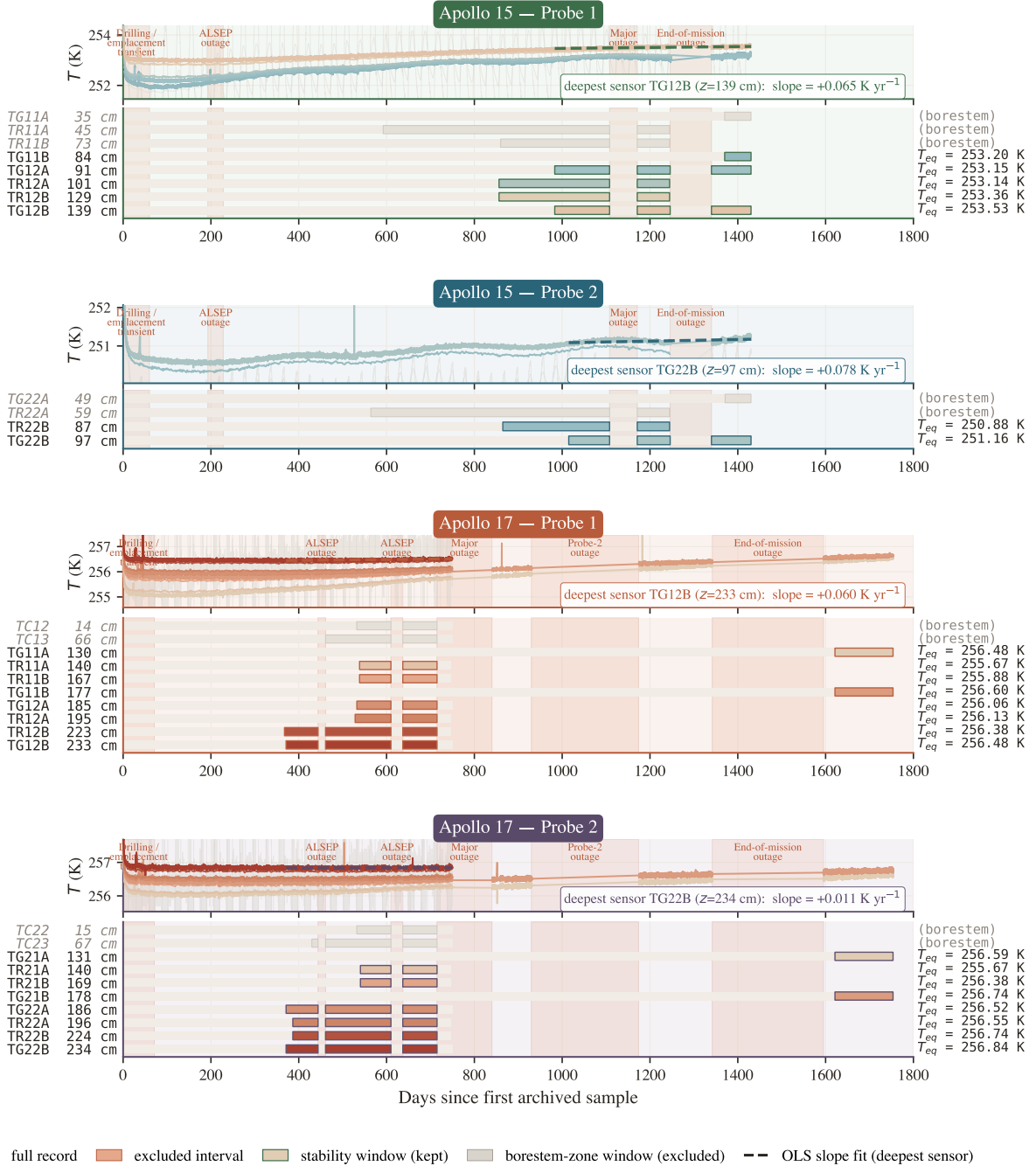
**Figure 2.** Apollo Heat-Flow Experiment landing-site context on the Clementine global albedo mosaic (USGS/NASA, public domain). **(a)** Global equirectangular map showing the two HFE boreholes; dashed coloured rectangles mark the zoom regions of panels (b) and (c). **(b)** Apollo 15 regional zoom ( $\pm 15^\circ$  longitude,  $\pm 10^\circ$  latitude), Imbrium–Apennine boundary, mare/highland margin. **(c)** Apollo 17 regional zoom, basaltic embayment of Mare Serenitatis at the Taurus–Littrow highlands edge (Korotev, 2005; Wieczorek et al., 2006; Crawford, 2015). Borehole coordinates from the ALSEP gazetteer.

(Nagihara et al., 2018), and slow instrumental drift, is excluded. For each sensor we extract automatically the longest trailing segment that is statistically indistinguishable from a flat steady state and average within it.

For each sensor, candidate start points covering 55–80% of the record (13 equally-spaced candidates; those retaining less than 20% of the record are excluded by a minimum-tail guard, capping the effective range at 80%) are evaluated by ordinary-least-squares linear fit to all temperatures between the candidate point and the end of the record. The accepted *stability window* is the earliest candidate for which the linear-fit slope satisfies  $|d\langle T \rangle / dt| < 0.08 \text{ K/yr}$  ( $\approx 2.2 \times 10^{-4} \text{ K/d}$ ): the longest tail consistent with thermal equilibrium at this stringency. The adopted threshold is approximately  $4.6\times$  stricter than the  $10^{-3} \text{ K/d}$  value sometimes used in HFE analyses, and was chosen to suppress bias from residual post-disturbance drift. If no candidate satisfies the criterion, the last 25% of the record is used as a fallback. We report the window-mean equilibrium temperature  $T_{\text{eq}} = \langle T(z) \rangle_{\Delta t}$  together with its within-window standard deviation, which serves as the per-sensor measurement uncertainty in downstream statistics. Sensors with central depth  $z < 80 \text{ cm}$  (the borestem-contaminated zone; §2.4) are plotted but excluded from the RMSE. Figure 3 shows the full per-probe temperature record at each site, with the documented disturbance intervals shaded

(Nagihara et al., 2018), the selected stability windows marked, and the fitted long-term drift  $d\langle T \rangle / dt$  reported per sensor.

The retrieval is insensitive to the precise threshold. Sweeping the threshold over 0.04–0.16 K/yr (a factor of four around the adopted value) shifts  $K_d^*$  by at most  $0.030.02$  mW/(m K) at A15 ( $0.70.4\%$  of nominal) and  $0.110.05$  mW/(m K) at A17 ( $1.00.6\%$ ), in both cases an order of magnitude below the bootstrap  $1\sigma$  uncertainty (§3.2). The adopted threshold therefore lies within the flat band of the threshold response.



**Figure 3.** Per-probe stability-window timeline at Apollo 15 and 17 (4 probes). Each probe panel shows, above, the raw HFE traces with coral bands marking documented data-quality events and the selected stability windows outlined; and, below, a depth-coded per-sensor Gantt strip broken at outages. Selection criteria and the role of the window-mean  $T_{eq}$  are described in §2.



## 2.2 Surface solar geometry and incident insolation

Each Apollo–Unix timestamp is propagated through SPICE (Acton 1996, 2018) using the JPL DE440 planetary ephemeris and the IAU-defined mean-Earth/polar-axis lunar body-fixed reference frame, with light-time and stellar-aberration corrections applied, to obtain the solar zenith angle  $\theta_{\odot}(t)$  at each site. The solver is forced with an idealized synodic insolation cycle: the daytime solar zenith angle at each site is approximated by  $\cos \theta_{\odot}(t) = \cos \phi \cos(2\pi t/P)$ , with  $\phi$  the site latitude and  $P = 29.531$  d the mean synodic period. Because the retrieval compares stability-window *mean* temperatures (§2) rather than time-resolved phases, ephemeris-level timing corrections do not enter the fit. The neglected solar declination ( $\pm 1.54^\circ$ ) and orbital eccentricity ( $\pm 3.3\%$  peak-to-peak in instantaneous flux,  $< 0.1\%$  in the cycle mean) perturb the mean insolation at or below the percent level, comparable to and subsumed by the Bond-albedo sensitivity of Table 4. The Moon has no atmosphere, so the solar irradiance incident on the local horizontal surface is

$$F_{\odot}(t) = \max(0, S_0 \cos \theta_{\odot}(t)), \quad (1)$$

with  $S_0 = 1361 \text{ W/m}^2$  (Kopp and Lean, 2011).

## 2.3 1-D solver and Hayne et al. (2017) thermophysics

The depth coordinate  $z$  is defined positive downward, with the regolith surface at  $z = 0$  and the lower boundary of the modelled column at  $z = z_{\max} = 5$  m. Conservation of energy in one dimension gives the heat equation

$$\rho(z) c_p(T) \frac{\partial T}{\partial t} = \frac{\partial}{\partial z} \left[ K(T, z) \frac{\partial T}{\partial z} \right]. \quad (2)$$

At the upper boundary the skin temperature  $T_s$  satisfies radiative balance between absorbed insolation, emitted thermal radiation, and the conductive flux into the column,

$$(1 - A) F_{\odot}(t) - \varepsilon \sigma T_s^4 - (-K \partial_z T)_{z=0} = 0, \quad (3)$$

with  $\sigma = 5.6704 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$  the Stefan–Boltzmann constant; equation (3) is solved iteratively for  $T_s$  at each time step. At the lower boundary a constant geothermal flux is imposed,

$$-K \partial_z T|_{z_{\max}} = Q_b. \quad (4)$$

The PDE is discretised on a geometric grid ( $\Delta z_0 = 2$  mm, growth ratio 1.08) and integrated with a Crank–Nicolson scheme at  $\Delta t = 1$  h, with a spin-up of 30 lunations to  $\max |\Delta T| < 5 \times 10^{-2} \text{ K}$ . The periodic steady state is computed by a flux-anchored outer iteration rather than a fixed-length spin-up: short multi-lunation integrations converge the diurnal skin, after which the sub-skin mean profile is reconstructed from the steady-state mean-flux closure  $d\langle T \rangle/dz = [Q_b - u_{\text{rect}}(z)]/K(\langle T \rangle, z)$ , where the rectified (eddy-correlation) flux  $u_{\text{rect}} = \langle K \partial_z T \rangle - K(\langle T \rangle) \partial_z \langle T \rangle$  is measured from the resolved cycle; the two steps are alternated until the anchor temperature at  $z = 0.55$  m is stable to  $5 \times 10^{-3} \text{ K}$ . This construction is required because a flux-bottom-boundary column relaxes on the diffusive timescale of the full 5-m domain ( $\sim 10^3$  lunations), so no fixed-length spin-up of practical cost is free of its initial condition. Convergence is certified rather than assumed: initial temperature guesses of 240 and 260 K yield mean profiles agreeing to  $\leq 0.03 \text{ K}$  everywhere in the column, and a 120-lunation verification integration started from the converged profile drifts by  $< 0.09 \text{ K}$  at the sensor depths, bounding the solver systematic on  $K_d^*$  at  $\lesssim 0.15 \text{ mW m}^{-1} \text{ K}^{-1}$  — an order of magnitude below the bootstrap uncertainty (§3.2). The Hayne et al. (2017) thermophysical inputs are adopted as given in Appendix A of the original paper:

$$K(T, z) = \left\{ K_s + (K_d - K_s) [1 - e^{-z/H}] \right\} \cdot \left[ 1 + \chi(T/T_{\text{ref}})^3 \right], \quad (5)$$

$$\rho(z) = \rho_d - (\rho_d - \rho_s) e^{-z/H}, \quad (6)$$

$$c_p(T) = c_0 + c_1 T + c_2 T^2 + c_3 T^3 + c_4 T^4. \quad (7)$$

**Table 1. Fixed inputs to the 1-D heat solver.** Three blocks: (A) the Hayne et al. (2017) conductivity-form coefficients (eq. 5), with  $K_d$  the single parameter retrieved in §2.5; (B) the four published coefficients of the Martínez and Siegler (2021)  $K(T, \rho)$  comparison model (eq. 8, forward only in the headline table; see also §4.3); (C) site-specific measured quantities shared by both conductivity models. The shared  $\rho(z)$  uses the Hayne-form profile (eq. 6); the specific heat is the Hemingway et al. (1973) polynomial  $c_p(T) = c_0 + c_1T + c_2T^2 + c_3T^3 + c_4T^4$  with  $(c_0, \dots, c_4) = (-3.61, 2.74, 2.36 \times 10^{-3}, -1.23 \times 10^{-5}, 8.91 \times 10^{-9})$ .

| Parameter                                                                                                                      | Symbol           | Value                              | Unit                                                    | Source                                            |
|--------------------------------------------------------------------------------------------------------------------------------|------------------|------------------------------------|---------------------------------------------------------|---------------------------------------------------|
| <i>(A) Hayne et al. (2017) smooth-exponential conductivity, all values from Hayne et al. (2017) App. A unless noted</i>        |                  |                                    |                                                         |                                                   |
| Surface conductivity                                                                                                           | $K_s$            | $7.4 \times 10^{-4}$               | $\text{W m}^{-1} \text{K}^{-1}$                         | <b>this work</b>                                  |
| Deep conductivity (global)                                                                                                     | $K_d$ (publ.)    | $3.4 \times 10^{-3}$               | $\text{W m}^{-1} \text{K}^{-1}$                         |                                                   |
| Deep conductivity (per-site)                                                                                                   | $K_d^*$          | §3.2                               | $\text{W m}^{-1} \text{K}^{-1}$                         |                                                   |
| $K(z)$ transition scale                                                                                                        | $H$              | 0.06                               | m                                                       |                                                   |
| Radiative coefficient                                                                                                          | $\chi$           | 2.7                                | —                                                       |                                                   |
| Radiative normalisation                                                                                                        | $T_{\text{ref}}$ | 350                                | K                                                       |                                                   |
| <i>(B) Martínez &amp; Siegler (2021) <math>K(T, \rho)</math> (forward only), coefficients from Martínez and Siegler (2021)</i> |                  |                                    |                                                         |                                                   |
| Contact, $\rho$ -slope                                                                                                         | $A_1$            | $5.082 \times 10^{-6}$             | $\text{W m}^{-1} \text{K}^{-1} (\text{kg m}^{-3})^{-1}$ |                                                   |
| Contact, intercept                                                                                                             | $A_2$            | $-5.10 \times 10^{-3}$             | $\text{W m}^{-1} \text{K}^{-1}$                         |                                                   |
| Radiative, $\rho$ -slope                                                                                                       | $B_1$            | $2.002 \times 10^{-13}$            | $\text{W m}^{-1} \text{K}^{-4} (\text{kg m}^{-3})^{-1}$ |                                                   |
| Radiative, intercept                                                                                                           | $B_2$            | $-1.953 \times 10^{-10}$           | $\text{W m}^{-1} \text{K}^{-4}$                         |                                                   |
| <i>Shared by both conductivity models</i>                                                                                      |                  |                                    |                                                         |                                                   |
| Surface bulk density                                                                                                           | $\rho_s$         | 1100                               | $\text{kg m}^{-3}$                                      | Hayne et al. (2017)                               |
| Deep bulk density                                                                                                              | $\rho_d$         | 1800                               | $\text{kg m}^{-3}$                                      | Hayne et al. (2017)                               |
| Specific heat                                                                                                                  | $c_p(T)$         | polyn. (caption)                   | $\text{J kg}^{-1} \text{K}^{-1}$                        | Hemingway et al. (1973)                           |
| <i>(C) Site-specific measured inputs (A15/A17)</i>                                                                             |                  |                                    |                                                         |                                                   |
| Latitude                                                                                                                       | $\phi$           | $26.1^\circ / 20.2^\circ \text{N}$ | —                                                       | ALSEP gazetteer                                   |
| Bond albedo                                                                                                                    | $A$              | 0.131 / 0.137                      | —                                                       | Vasavada et al. (2012)                            |
| Broadband emissivity                                                                                                           | $\epsilon$       | 0.95                               | —                                                       | Bandfield et al. (2015)                           |
| Basal heat flux                                                                                                                | $Q_b$            | 21 / 15                            | $\text{mW m}^{-2}$                                      | Langseth et al. (1976);<br>Nagihara et al. (2018) |
| Solar constant                                                                                                                 | $S_0$            | 1361                               | $\text{W m}^{-2}$                                       | Kopp and Lean (2011)                              |

Numerical values and sources for every coefficient, together with the site-specific measured quantities (latitude, albedo, emissivity, basal heat flux), are listed in Table 1. The basal heat flux reanalyses of Saito et al. (2007); Nagihara et al. (2018) are folded into the sensitivity sweep of §3.2.

As an independent comparison we also evaluate the Martínez and Siegler (2021) regolith conductivity model. Where Hayne et al. (2017) prescribes a depth-dependent envelope through  $K(T, z) = [K_s + (K_d - K_s)(1 - e^{-z/H})](1 + \chi(T/T_{\text{ref}})^3)$ , Martínez and Siegler (2021) replaces that envelope with an explicitly density-dependent two-mechanism decomposition,

$$K(T, \rho) = \underbrace{(A_1\rho + A_2)k_{\text{am}}(T)}_{\text{contact (phonon)}} + \underbrace{(B_1\rho + B_2)T^3}_{\text{radiative}}. \quad (8)$$

The first term is the contact (phonon) conductivity transmitted through the network of grain-to-grain contacts; it carries the temperature dependence of an amorphous solid through the low-temperature polynomial  $k_{\text{am}}(T)$  of Martínez and Siegler (2021) and scales linearly with the local bulk density, reflecting the increased number of grain-to-grain contacts per unit volume at higher packing. The second term is the radiative contribution from infrared photon exchange across pore voids; it scales as  $T^3$  (the derivative of the Stefan–Boltzmann emission with respect to temperature, characteristic of photon-mediated transport) and is density-modulated with the opposite physical content, since denser packing reduces the void fraction available for radiative exchange. The four coefficients  $(A_1, A_2, B_1, B_2)$  are calibrated globally against Diviner brightness temperatures and are listed in Table 1. The density profile  $\rho(z)$  used in this comparison is the same Hayne-form exponential used in the Hayne column, so the two models differ in their  $K$  formulation alone.

The Martínez and Siegler (2021) formulation has no single  $K_d$  analogous to the parameter

retrieved under the Hayne form: both terms in equation (8) vary continuously with depth through  $\rho(z)$  and with temperature through  $T(z)$ . The published model contains no free parameter, so its evaluation at each site yields a single forward prediction (Table 2). To diagnose, in the discussion (§4.3), whether the published Martínez and Siegler (2021) form could fit both sites if it were granted one per-site lever analogous to the Hayne  $K_d$ , we additionally perform an alternative retrieval in which the ~~deep~~-asymptote of the Hayne-form density profile entering equation (8) is rescaled by a single per-site scalar  $\alpha$ ,

$$\rho_d^{\text{site}} = \alpha \cdot 1800 \text{ kg m}^{-3}, \quad (9)$$

with  $\rho_s = 1100 \text{ kg m}^{-3}$ , the e-folding depth  $H$ , and the published coefficients ( $A_1, A_2, B_1, B_2$ ) and  $k_{\text{am}}(T)$  held fixed;  $\alpha = 1$  recovers the published forward prediction. The scalar  $\alpha$  is our diagnostic construction (the Martínez and Siegler (2021) model itself is not refit, only re-evaluated at a per-site density), and its physical interpretation is the per-site ~~deep~~-asymptotic bulk density that the published  $K(T, \rho)$  would require to reproduce the in-situ data. The  $\alpha$  retrieval is not part of the headline  $K_d$  result; it is interpreted in §4.3.

## 2.4 Borestem diagnostic and the meter-scale-sensor cut

The ~~deep~~-meter-scale ( $z > 80 \text{ cm}$ ) sensor set used in the  $K_d$  retrieval is defined by a physics-based amplitude diagnostic, not by the residuals it would produce. The amplitude of the diurnal wave serves as a frequency-domain check independent of the annual-mean comparison. The per-sensor diurnal amplitude is ~~computed as the half-range of the local solar time folded median temperature estimated as  $\sqrt{2} \sigma_{\Delta t}$ , where  $\sigma_{\Delta t}$  is the within-stability-window standard deviation; the proxy equals the half-range exactly for a sinusoidal signal, is diurnal-dominated for shallow sensors, and saturates at the digitisation noise floor below the skin depth — precisely the contrast the diagnostic requires~~ (Fig. 4). Two features are diagnostic. Above  $z = 80 \text{ cm}$ , observed amplitudes exceed the modelled regolith-attenuation curve by an order of magnitude or more, the signature of heat conducted along the fibreglass borestem from the lunar surface, which thermally short-circuits the upper  $\sim 80 \text{ cm}$  of regolith (Nagihara et al., 2018). Below  $80 \text{ cm}$ , both published  $K(z)$  forms predict diurnal amplitudes  $\lesssim 10^{-7} \text{ K}$  ( $\sim 18$  e-foldings of the diurnal skin depth  $\delta \sim 3\text{--}5 \text{ cm}$ ), two to three orders of magnitude below the  $0.05\text{--}0.2 \text{ K}$  digitisation noise floor of the restored record. ~~Deep-sensor-Meter-scale-sensor~~ amplitudes therefore do not constrain the thermal diffusivity  $\kappa = K/(\rho c_p)$  at the in-situ scale; the Hayne and Martínez and Siegler (2021) attenuation curves plotted in Figure 4 agree to within  $\sim 10\%$  in skin depth and overlap visually below the noise floor, both indicating “deep = unresolved”. The shallow-sensor amplitude excess, evaluated against a prediction that depends only on  $K(T, z)$ ,  $\rho(z)$ , and  $c_p(T)$ , defines the ~~deep~~-meter-scale ( $z > 80 \text{ cm}$ ) validation set used in the retrieval.

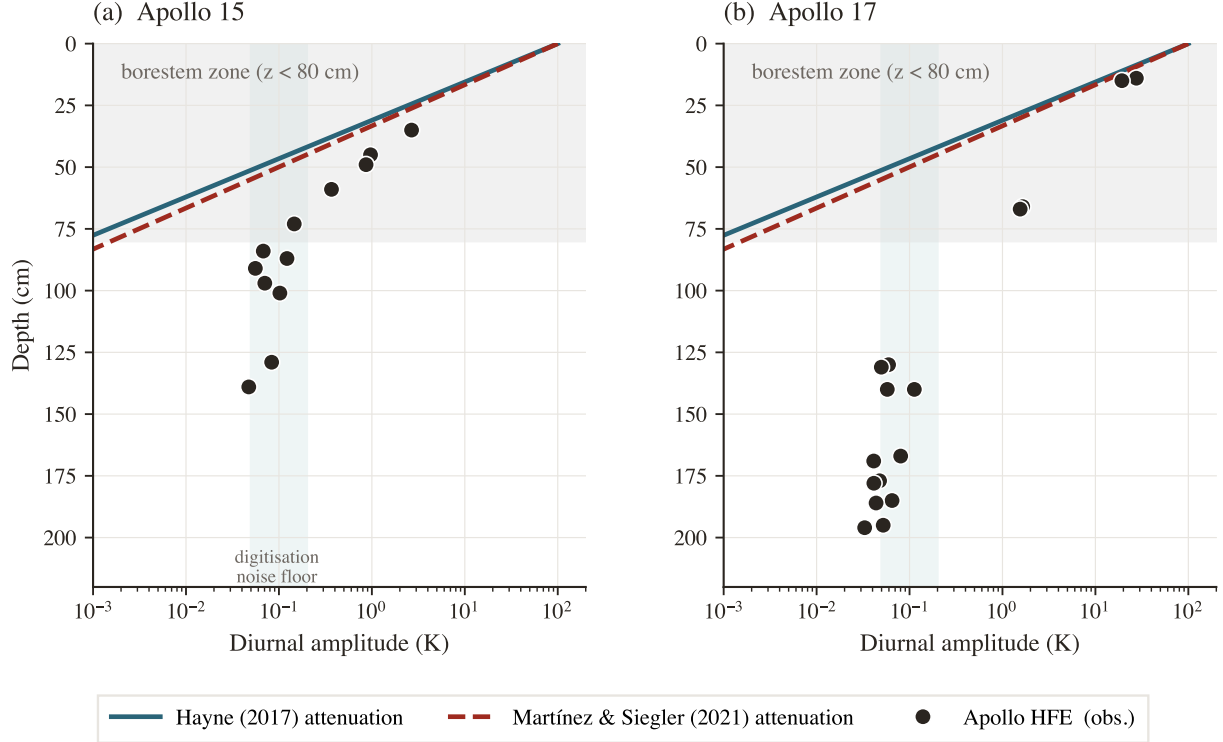
## 2.5 Per-site $K_d$ retrieval (Hayne shape held fixed)

$K_d$  is retrieved separately at each site by minimising the ~~deep-sensor-meter-scale-sensor~~ RMSE while holding the remainder of the Hayne functional form (eqs. 5–7) fixed. ( $K_s, H, \chi, T_{\text{ref}}$ ) and the density and  $c_p$  functions are held at their published values, and  $K_d$  is swept across site-specific grids: ~~20 points spanning  $1.5\text{--}9.0 \times 10^{-3} \text{ W/(m K)}$  at A15 and 24 points spanning  $3.0\text{--}18.0 \times 10^{-3} \text{ W/(m K)}$  at A17. The upper end of the A17 grid bounds the parabolic minimum; the bootstrap upper tail (Fig. 7) extends slightly beyond the swept range through parabolic extrapolation.~~ 28 points spanning  $1.0\text{--}15.0 \times 10^{-3} \text{ W/(m K)}$  at A15 and 30 points spanning  $3.0\text{--}25.0 \times 10^{-3} \text{ W/(m K)}$  at A17, sized so that both parabolic minima and the bootstrap upper percentiles fall inside the swept range (Fig. 7). At each  $K_d$  a full Crank–Nicolson spin-up is run to convergence, and the ~~deep-sensor-meter-scale-sensor~~ RMSE is evaluated as

$$\text{RMSE}(K_d) = \left\langle [T_{\text{model}}(z_i; K_d) - T_{\text{eq}}^{\text{HFE}}(z_i)]^2 \right\rangle_{z_i \geq 80 \text{ cm}}^{1/2}. \quad (10)$$

The point estimate  $K_d^*$  is the parabolic minimum through the three grid values bracketing  $\arg \min \text{RMSE}$ . Its uncertainty is quantified by a non-parametric bootstrap with  $N_{\text{boot}} = 1500$  resamples of the ~~deep-sensor-meter-scale-sensor~~ set drawn with replacement, with the  $\pm 2.5 \text{ cm}$  sensor-placement uncertainty (Nagihara et al., 2018) propagated as a per-resample Gaussian depth jitter, following





**Figure 4.** Diurnal amplitude (half-range of the local-solar-time-folded median temperature estimated as  $\sqrt{2} \times$  the within-stability-window standard deviation, the half-range of a sinusoidal signal) versus sensor depth at both Apollo sites. Black markers: observed; solid teal: Hayne et al. (2017) regolith attenuation; dashed red: Martínez and Siegler (2021) attenuation, both evaluated at  $T = 250$  K with the published deep-regolith meter-scale regolith density of  $1800 \text{ kg m}^{-3}$  and  $c_p(250 \text{ K}) \approx 670 \text{ J kg}^{-1} \text{ K}^{-1}$ , the Hayne form including its radiative conductivity term. The two model curves differ by  $\sim 10\%$  in predicted skin depth and are visually indistinguishable on the log axis at all relevant depths. The shaded band is the borestem zone ( $z < 80$  cm). The order-of-magnitude shallow-sensor excess is the borestem heat-short signature.

standard small-sample regression practice (Press et al., 2007). The bootstrap distribution defines 95% confidence intervals on  $K_d^*$  and on the inter-site contrast  $\Delta K_d^*$ , and provides the null-hypothesis test for inter-site equality (Fig. 7). Percentile-method intervals are reported throughout; bias-corrected accelerated (BCa) intervals (Efron and Tibshirani, 1993) agree with the percentile endpoints to within  $1.3 \text{ mW}/(\text{m K})$  at both sites and on the contrast, and do not alter any qualitative conclusion. The retrieval has one free parameter.

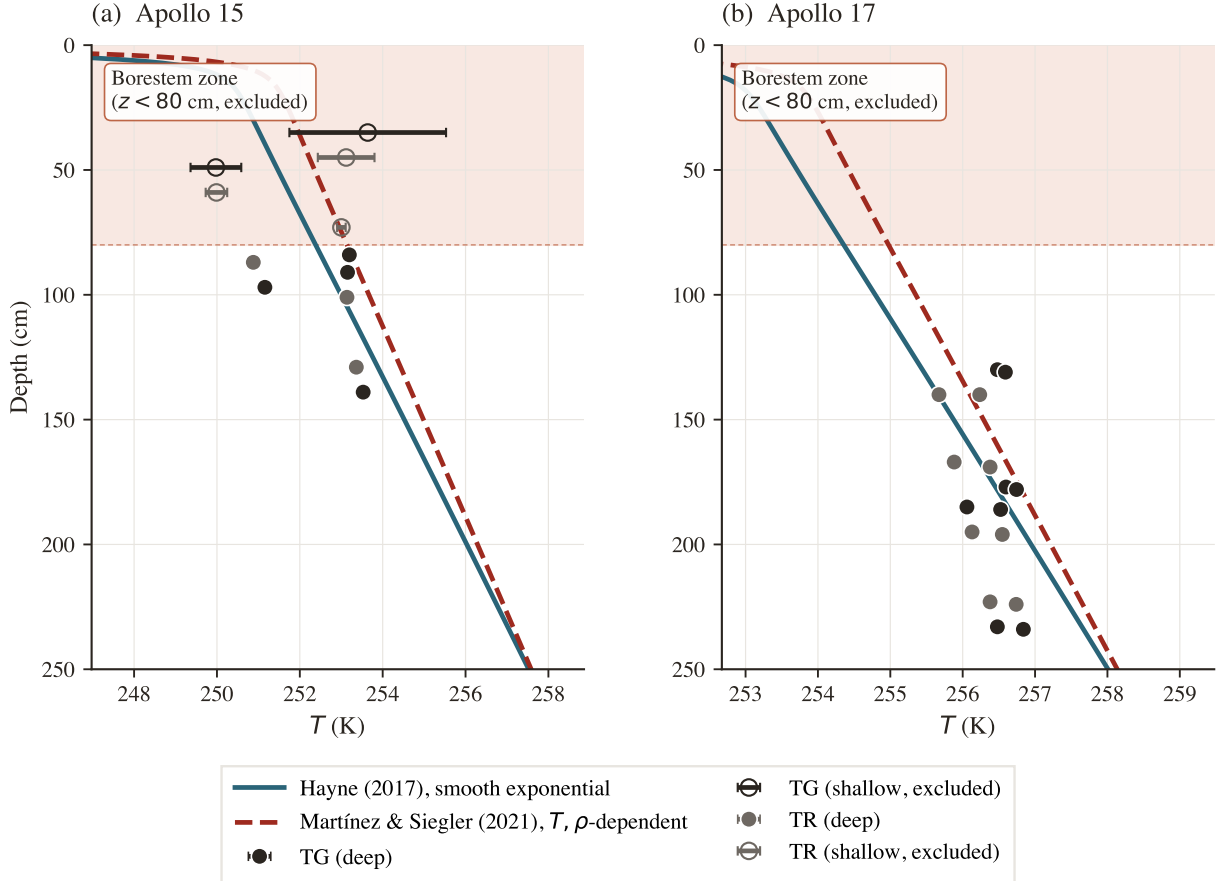
Models with different free-parameter counts on the same deep-sensor-meter-scale-sensor set are compared using the small-sample-corrected Akaike information criterion (Burnham and Anderson, 2002),  $\text{AICc} = N \ln(\text{RSS}/N) + 2k + 2k(k+1)/(N-k-1)$ , where RSS is the deep-sensor-meter-scale-sensor residual sum of squares and  $k$  is the number of fitted parameters ( $k = 0$  for fixed- $K_d$  rows,  $k = 1$  for each retrieval). Differences  $\Delta \text{AICc}$  are reported relative to the best model at each site; following Burnham and Anderson (2002),  $\Delta \text{AICc} \lesssim 2$  indicates statistically indistinguishable models and  $\Delta \text{AICc} \gtrsim 10$  a decisive preference. A reduced  $\chi^2$  is not reported: the within-stability-window scatter ( $\sigma \sim 0.03\text{--}0.05 \text{ K}$ ) characterises short-term thermometer precision rather than the model-adequacy error budget, yields  $\chi_\nu^2 \gg 1$  for every model, and is therefore uninformative for model selection in this context.

### 3 Results

#### 3.1 Annual-mean subsurface profile and the global-model gap

Neither published global model reproduces both Apollo boreholes without per-site adjustment. Both published global models reproduce the annual-mean borehole profiles to within  $\sim 1 \text{ K}$ , but systematic per-site residuals remain. Figure 5 compares the two modelled time-averaged  $\langle T(z) \rangle$  profiles, Hayne

et al. (2017) at the global  $K_d = 3.4 \text{ mW}/(\text{m K})$  and Martínez and Siegler (2021)  $K(T, \rho)$  evaluated forward, against the stability-window means. The published Hayne value runs warm relative to the HFE record by a mean of  $+1.03 \text{ K}$  at A15 and  $+2.88 \text{ K}$  at A17  $+0.49 \text{ K}$  at A15 and  $+0.16 \text{ K}$  at A17, with RMSE 1.03 and 0.69 K — the residual structure, not the mean offset, carries the per-site information; the Martínez and Siegler (2021) prediction is closer at both sites but underpredicts A17 by  $\sim 2 \text{ K}$  is comparable (RMSE 0.99 and 0.64 K) (Table 2). The shaded band ( $z < 80 \text{ cm}$ ) marks the borestem-contaminated zone, excluded *a priori* from the quantitative comparison (§2.4). The residual that remains motivates structure that remains — in particular the under-predicted temperature gradient between the 80-cm and deepest sensors at A17 — motivates the per-site  $K_d$  retrieval of §3.2.



**Figure 5.** Annual-mean subsurface temperature  $\langle T(z) \rangle$  for the two published conductivity models at their nominal (un-retrieved) parameters, Hayne et al. (2017) smooth exponential at the global  $K_d = 3.4 \text{ mW}/(\text{m K})$  (solid), and the Martínez and Siegler (2021) temperature- and density-dependent  $K(T, \rho)$  forward (dashed), against the Apollo HFE stability-window means. Markers are coded by sensor type (charcoal = TG, grey = TR); horizontal bars are the within-window standard deviation. The shaded band is the borestem-contaminated zone ( $z < 80 \text{ cm}$ ), excluded from the RMSE.

### 3.2 Per-site $K_d$ retrieval

The deep-sensor-meter-scale-sensor RMSE exhibits a well-resolved minimum at each site:

$$K_{d,A15}^* = 4.86^{+1.12}_{-0.54} [3.94, 14.36] 4.58^{+1.33}_{-0.28} [4.12, 7.45] \text{ mW}/(\text{m K}), \quad K_{d,A17}^* = 11.23^{+8.12}_{-0.77} [10.18, 21.62] 8.12^{+0.49}_{-0.61} [6.12, 10.12] \text{ mW}/(\text{m K}),$$

with asymmetric  $1\sigma$  bootstrap uncertainties and 95% confidence intervals in brackets (Fig. 7). The deep-sensor-meter-scale-sensor RMSE decreases from  $1.37 \text{ K}$  and  $2.96 \text{ K}$   $1.03 \text{ K}$  and  $0.69 \text{ K}$  under the published global  $K_d$  to  $0.91 \text{ K}$  and  $0.30 \text{ K}$   $0.95 \text{ K}$  and  $0.34 \text{ K}$  under the per-site retrieval ( $N = 7$  at A15,  $N = 16$  at A17; Table 2), and the residual bias falls to  $\leq 0.03 \text{ K}$ ; at A17 the refit halves the RMSE and removes the gradient misfit (bias  $+0.04 \text{ K}$ ), while at A15 the improvement is modest

**Table 2. Headline ~~deep-sensor-meter-scale-sensor~~ RMSE table.** ~~Deep-sensor-Meter-scale-sensor~~ ( $z > 80$ – $z = 80$ – $234$  cm) validation statistics for the three forward configurations confronted with the same Apollo HFE record: the Hayne et al. (2017) form at its published global  $K_d$ , the Hayne et al. (2017) form with  $K_d$  refit per site (the retrieval of §3.2), and the Martínez and Siegler (2021)  $K(T, \rho)$  model evaluated forward at each site.  $K_d$  in  $\text{mW m}^{-1} \text{K}^{-1}$  (95% bootstrap CI in brackets for the Hayne site-fit row). Bias =  $\text{mean}(\text{model} - \text{obs})$ . A diagnostic per-site retrieval under the Martínez and Siegler (2021) form is presented separately in §4.3.

| Site      | Model                               | Fitted knob                 | $N$ | RMSE (K)    | Bias (K) |
|-----------|-------------------------------------|-----------------------------|-----|-------------|----------|
| Apollo 15 | Hayne (2017), global $K_d$          | —                           | 7   | <b>1.03</b> | +0.49    |
| Apollo 15 | Hayne, $K_d^*$ fit                  | $K_d^* = 4.57 [4.12, 7.45]$ | 7   | <b>0.95</b> | +0.32    |
| Apollo 15 | Martínez and Siegler (2021) forward | —                           | 7   | <b>0.99</b> | +0.41    |
| Apollo 17 | Hayne (2017), global $K_d$          | —                           | 16  | <b>0.69</b> | +0.16    |
| Apollo 17 | Hayne, $K_d^*$ fit                  | $K_d^* = 8.12 [6.85, 9.04]$ | 16  | <b>0.34</b> | +0.04    |
| Apollo 17 | Martínez and Siegler (2021) forward | —                           | 16  | <b>0.64</b> | −0.27    |

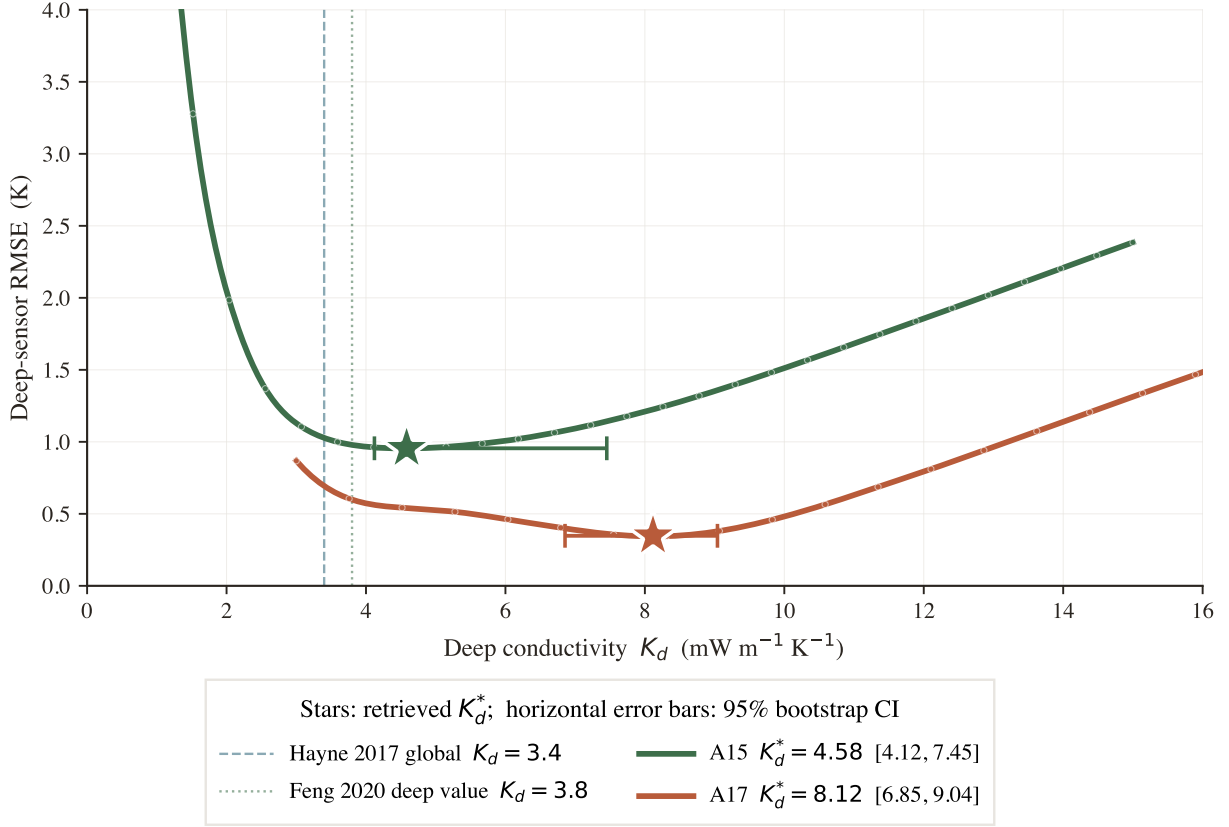
and the global value is not excluded by the RMSE criterion alone — it is excluded by the bootstrap CI, whose lower bound (4.12) lies above 3.4.  $K_d$  is therefore the dominant identifiable lever between the Hayne form and the in-situ HFE record once the borestem zone is excluded, with the caveat that the per-site evidence is decisive at A17 ( $\Delta\text{AICc} \approx 19$  against the global value) but not at A15 alone ( $\Delta\text{AICc} \approx 3$  in favour of the *global* value once the parameter penalty is paid; the inter-site contrast, not the A15 refit, carries the *statistical* result).

The minimum is not an artefact of the sweep grid. Extending the sweep beyond the fitting envelope reveals that each RMSE curve rises monotonically on the high- $K_d$  side (Fig. 6), confirming bracketed minima rather than descending shoulders. ~~A fivefold refinement of the grid around each minimum, combined with a factor-of-two tightening of the spin-up length and convergence tolerance (60 lunations to  $5 \times 10^{-3}$  K versus the production 30 lunations to  $5 \times 10^{-2}$  K), shifts  $K_d^*$  by less than  $0.10 \text{ mW}/(\text{m K})$  at both sites, well within the bootstrap uncertainty quoted above.~~ Nor is it an artefact of the thermal-state initialisation: the equilibrium construction of §2.3 is certified independent of its initial guess to  $\leq 0.03$  K, which bounds the solver contribution to  $K_d^*$  at  $\lesssim 0.15 \text{ mW}/(\text{m K})$ , an order of magnitude below the bootstrap uncertainty quoted above.

The Martínez and Siegler (2021) model, evaluated forward at each site with its published global coefficients, serves as the global-baseline comparison: ~~deep-sensor-meter-scale-sensor~~ RMSE of ~~1.10 K at A15 and 2.47 K at A17, with mean biases of +0.6 K and +2.3 K~~ 0.99 K at A15 and 0.64 K at A17, with mean biases of +0.41 K and −0.27 K (Table 2). ~~It reproduces A15 to within a factor of  $\sim 1.2$  of the Hayne site-fit RMSE but underpredicts A17 by an amount comparable to the Hayne global miss, as expected for a globally calibrated  $K(T, \rho)$  without per-site freedom.~~ It reproduces both sites to within  $\sim 1$  K without any per-site freedom — comparable to the Hayne site-fit at A15, though still a factor of  $\sim 1.9$  above it at A17. A diagnostic per-site retrieval under this form is presented in §4.3, where the implications for the  $K(T, \rho)$  parameterization are discussed.

**Sensitivity to the borestem exclusion depth.** The retrieval is repeated with the borestem exclusion depth  $z_b$  swept over 60–90 cm. The retrieved values remain stable across all admissible cuts:  $K_{d,A15}^* = 4.7$  — ~~4.9~~ 4.6 — ~~−4.9~~ and  $K_{d,A17}^* = 11.28$  — ~~11.28~~ 11.1  $\text{mW}/(\text{m K})$  for  $z_b = 70, 80$ , and  $90$  cm, yielding a contrast of ~~6.3–6.5~~ 3.2–3.6  $\text{mW}/(\text{m K})$ , unchanged within rounding from the nominal value. The 60 cm cut alone deviates ( $K_{d,A17}^* = 18.89$  — ~~18.89~~ 18.1, contrast ~~14.24~~ 2.2), because at this depth a sensor inside the borestem-affected zone re-enters the A17 set and inflates the retrieved gradient, precisely the contamination that the amplitude-based cut (§2.4) is designed to exclude. The retrieval is therefore insensitive to  $z_b$  across its physically defensible range, and the 60 cm outlier confirms rather than undermines the borestem diagnostic.

Two summary statistics must be distinguished. The  $K_d^*$  values quoted above are parabolic point estimates of the RMSE minimum; the bootstrap medians of the same quantities are ~~4.88 and 11.51~~ 4.63 and 8.13  $\text{mW}/(\text{m K})$  (Table 4), ~~marginally higher because each bootstrap distribution is right-skewed.~~ The inter-site contrast is defined as the bootstrap median of the paired within-



**Figure 6.** Per-site  $K_d$  retrieval under the Hayne et al. (2017) functional form, with all other Hayne et al. (2017) parameters held at their published values. Deep-sensor-Meter-scale-sensor RMSE( $K_d$ ) at each site, extended past the fitting grid so each curve is shown rising again on the high- $K_d$  side: the minima are genuine bracketed bowls, not descending shoulders. Markers: forward-model evaluations; stars: retrieved  $K_d^*$ ; shaded band: 95% bootstrap CI; horizontal error bars: 95% bootstrap CI. Dashed vertical: Hayne et al. (2017) global  $K_d = 3.4$ ; dotted: the Feng et al. (2020) deep value  $K_d = 3.8$ .

resample difference,  $\Delta K_d^* = 6.63^{+7.43}_{-1.25} 3.31^{+0.71}_{-1.28}$  mW/(m K) ( $1\sigma$ , asymmetric; 95% CI  $[-3.2, 16.6]$ ,  $p \approx 0.05$  [0.37, 4.58],  $p \approx 0.011$  for the null of inter-site equality; Fig. 7). The paired construction preserves the resample-level correlation between sites and is therefore the statistically appropriate estimator; this is also the reason  $6.63 3.31$  differs slightly from the difference of the point estimates ( $11.23 - 4.86 = 6.37 8.12 - 4.58 = 3.53$ ). The broad A15 upper tail reflects a weakly constrained shoulder in RMSE( $K_d$ ) at A15 for  $N = 7$  deep-meter-scale sensors, even though the central parabolic minimum is firmly at  $4.86 4.58$ . The reported  $1\sigma$  intervals propagate sensor-set composition, the  $\pm 2.5$  cm depth-placement uncertainty (Nagihara et al., 2018), and the parabolic curve-fit error; component-by-component impacts of the remaining Hayne inputs are tabulated in Table 4. Both retrievals reject the global  $K_d = 3.4$  mW/(m K) of Hayne et al. (2017) at the 95% level (bootstrap CI lower bounds 4.12 and 6.85).

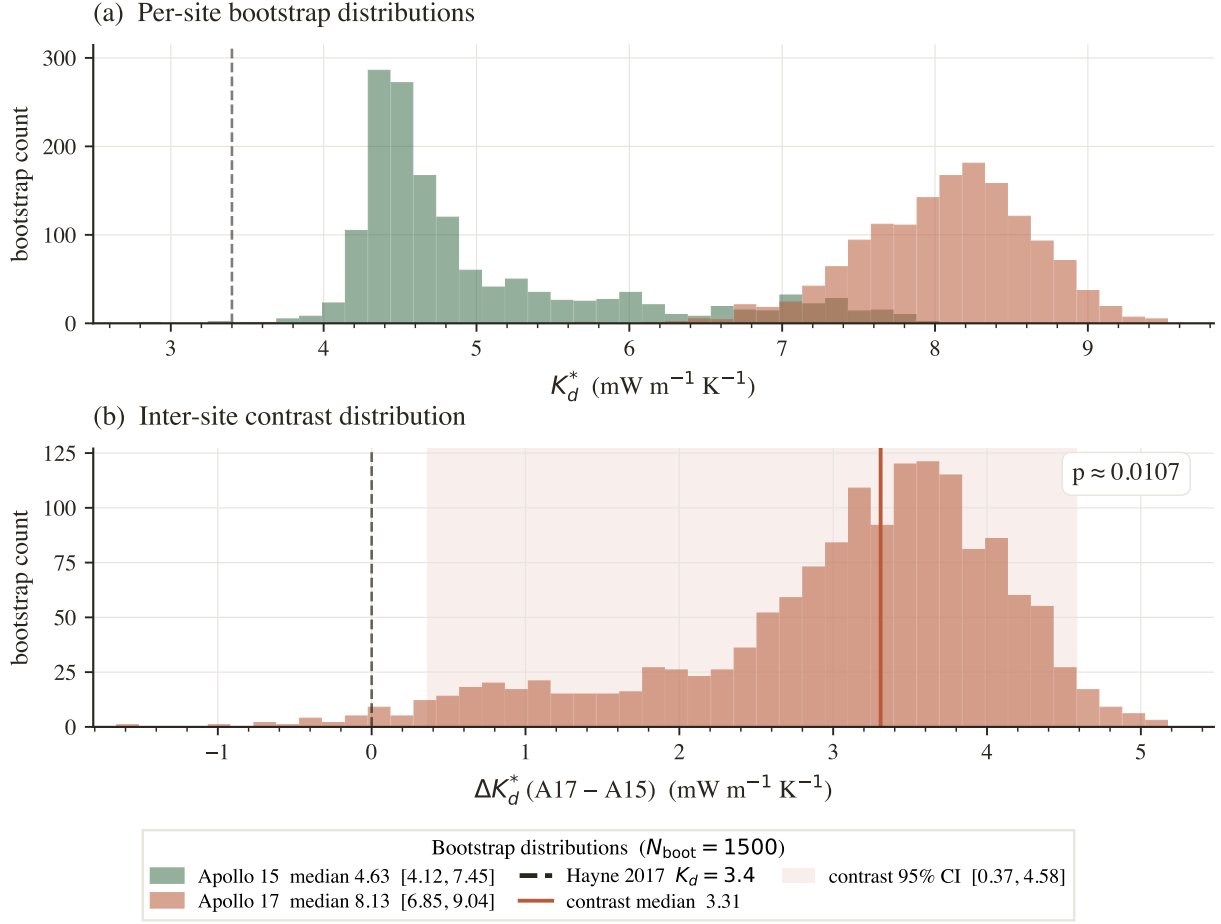
Applied at each site, the retrieved  $K_d^*$  values yield deep-sensor-meter-scale-sensor profiles that track the HFE observations within the  $\sim 1$  K spread of the data (Fig. 8). The Martínez and Siegler (2021) forward, plotted alongside, reproduces A15 to within a factor of  $\sim 1.2$  of the Hayne site-fit but underpredicts A17.

A full per-component error budget for  $K_d^*$  is deferred to §4.2 (Table 4), where the basal-flux degeneracy that dominates the budget is discussed.

### 3.3 Consistency checks

Three diagnostics test the per-site  $K_d^*$  values. Their status as independent or internal is identified explicitly.

An independent Diviner surface-temperature closure test (Fig. 11, discussed in §4.3) shows that



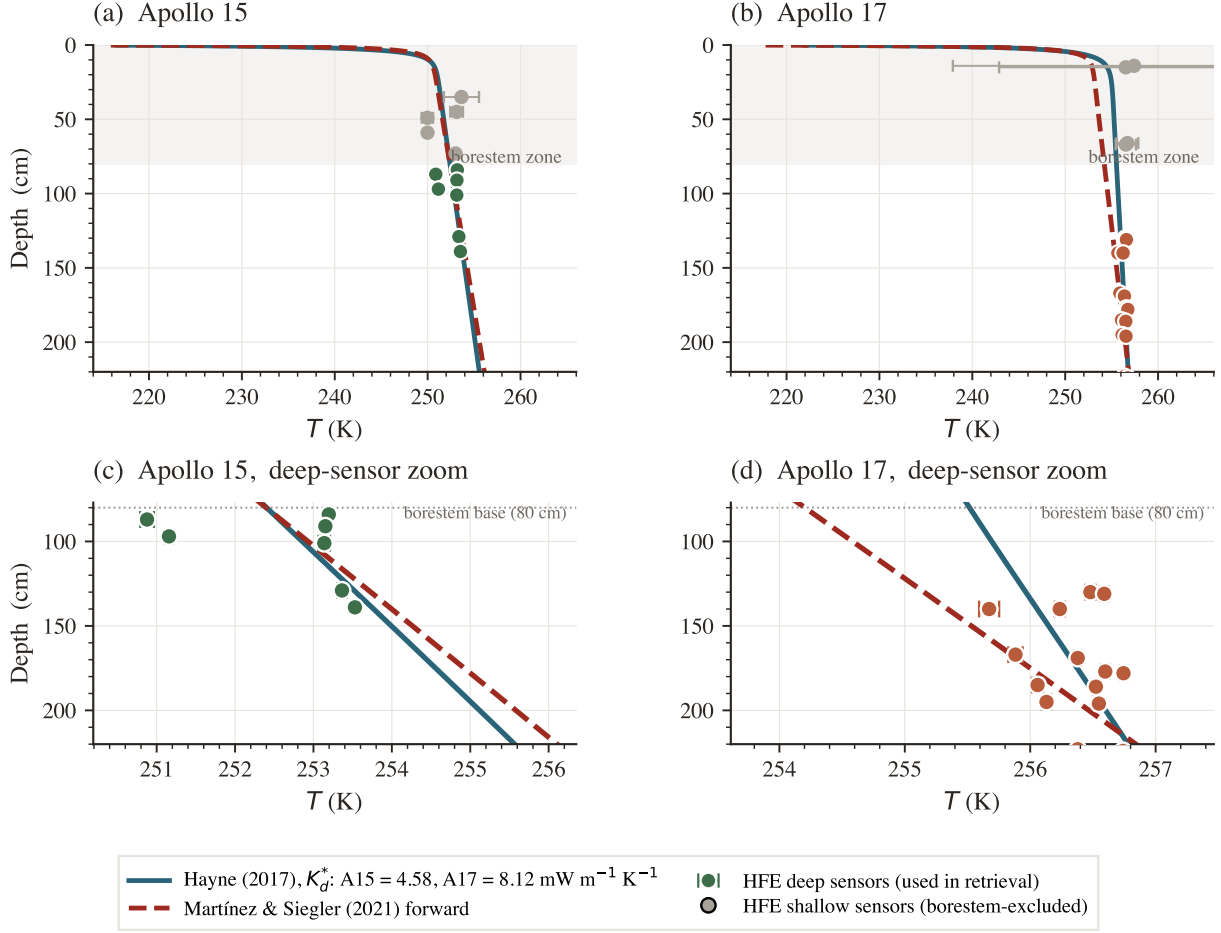
**Figure 7. Bootstrap distributions** ( $N_{\text{boot}} = 1500$ , **deep-sensor-meter-scale-sensor** resampling with  $\pm 2.5$  cm depth jitter). (a) Per-site  $K_d^*$  histograms with 95% CIs in the legend; dashed vertical at the Hayne et al. (2017) global value. (b) Inter-site contrast  $\Delta K_d^* = K_d^{A17} - K_d^{A15}$ ; 95% CI shaded, **lower bound just touching zero** ( $p \approx 0.05$ ) **lower bound above zero** ( $p \approx 0.011$ ).

both conductivity models reproduce the canonical lunar diurnal shape, confirming that the diurnal forcing chain is correctly captured. Two further internal checks follow.

(i) *TG vs. TR cross-prediction.* Fitting  $K_d$  on the gradient-bridge sensors (TG) alone and predicting the ring-bridge (TR) sensors, and vice versa, yields held-out RMSEs within  $\sim 0.25$  K of the in-sample values at both sites. (ii) *Leave-one-deepest-out.* Withholding the deepest sensor ( $z = 139$  cm at A15,  $z = 234$  cm at A17) and refitting  $K_d$  on the remainder predicts the held-out temperature to within  $|\Delta| = 0.31$  K and  $|\Delta| = 0.16$  K respectively. Both internal checks confirm that the retrieval is not driven by a single sensor or sensor type; neither constitutes an independent test of the inter-site contrast.

*Bayesian cross-check.* The non-parametric bootstrap is reported as the primary uncertainty estimate throughout because it makes no assumption about the basal heat flux and propagates only quantities measured at the boreholes (sensor-set composition and placement depth). As an independent cross-check on the *direction* of the contrast, not as a second estimate of its magnitude, we also ran a Bayesian inversion propagating the Saito et al. (2007); Nagihara et al. (2018) prior on  $Q_b$ . Its marginal MCMC medians are  $K_{d,A15} = 3.9$  and  $K_{d,A17} = 13.16$  mW m<sup>-1</sup> K<sup>-1</sup> (contrast median  $+9.2$ ). These differ from the bootstrap point estimates ( $4.86, 11.23$ ; contrast  $+6.64$ ,  $8.12$ ; contrast  $+3.3$ ) by construction, not by disagreement: the  $Q_b$  prior slides each marginal along the  $K_d/Q_b$  degeneracy ridge (§4.1), so the two methods are not independent measurements of the absolute  $K_d$ . **They agree on the ordering,  $P(K_d^{A17} > K_d^{A15}) \approx 96\%$ , since the degeneracy ridge shifts both sites together.** The ordering survives the  $Q_b$  smearing with  $P(K_d^{A17} > K_d^{A15}) \approx 83\%$ : conditional on the published basal fluxes the contrast is significant ( $p \approx 0.011$ ), but admitting





**Figure 8.** Annual-mean  $T(z)$  at the two sites under the two conductivity models, against the Apollo HFE ~~deep-meter-scale~~ sensors. Solid teal: Hayne et al. (2017) at the per-site retrieved  $K_d^*$ ; dashed red: Martínez and Siegler (2021) forward (no fitted knob). Top: full profile; bottom: ~~deep-sensor-meter-scale-sensor~~ zoom. Coloured markers: HFE ~~deep-meter-scale~~ sensors (used in retrieval); grey markers: borestem-affected shallow sensors (excluded).

the full reanalysis envelope on  $Q_b$  dilutes it — the honest statement is that the contrast in the meter-scale temperature gradient is established, while its attribution to conductivity rather than basal flux depends on the  $Q_b$  assumption (§4.2). The bootstrap interval on the contrast magnitude ( $\Delta K_d^* = 6.6$ , 95% CI  $[-3.2, 16.6]$   $\Delta K_d^* = 3.3$ , 95% CI  $[0.4, 4.6]$ ) is carried forward as the headline measurement.

## 4 Discussion

### 4.1 The two boreholes require different meter-scale temperature gradients

The HFE record does not admit a single ~~deep-meter-scale~~ conductivity that fits both sites. Under the Hayne form at the published basal heat fluxes, the value that zeroesfits the A17 ~~deep-sensor meter-scale-sensor~~ residual (~~11.238.12~~) overcorrects A15 to ~~-0.50 K~~ degrades A15 to a ~~+0.79 K~~ bias (RMSE 1.22 K), and the value that zeroesfits A15 (~~4.864.58~~) leaves A17 warm by ~~+1.63 K~~ degrades the A17 RMSE from 0.34 to 0.54 K. The published global value of Hayne et al. (2017), applied unchanged, runs warm at both sites but by significantly different amounts (~~+1.03 K at A15, +2.88 K at A17; Table 2~~) (~~+0.49 K at A15, +0.16 K at A17; Table 2~~) and, more importantly, underpredicts the A17 meter-scale temperature gradient. The per-site retrievals are  $K_d^{A15} = 4.864.58$  and  $K_d^{A17} = 11.238.12 \text{ mW}/(\text{m K})$  (95% bootstrap CIs ~~[3.94, 14.36]~~ and ~~[10.18, 21.62]~~ ~~[4.12, 7.45]~~ and ~~[6.85, 9.04]~~). The lower 95% bound on the contrast distribution (Fig. 7b) coincides with zero ( $p \approx 0.05$ ); the magnitude of the contrast is therefore not established at the 95% level by two

boreholes alone, but the *requirement* of different meter-scale temperature gradients is lies above zero ( $\Delta K_d^* = 3.31$ , 95% CI [0.37, 4.58],  $p \approx 0.011$ ): conditional on the published basal heat fluxes, both the *requirement* of different meter-scale temperature gradients and the *magnitude* of the conductivity contrast are established at the 95% level. The conditionality matters — under the  $K_d/Q_b$  degeneracy of §4.2 the attribution to conductivity rather than basal flux still rests on the  $Q_b$  assumption.

The pooled three-model comparison (Table 3) ~~confirms this conclusion directly~~ points the same way, more weakly. A per-site  $K_d$  with  $Q_b$  fixed at the published values (M1) is preferred over a shared  $K_d$  with per-site  $Q_b$  free within the published envelope (M2) ~~and~~,  $\Delta\text{AICc} = 4.0$  and over the strict null of shared  $K_d$  and fixed  $Q_b$  (M3) ~~by  $\Delta\text{AICc} \approx 5$~~ ,  $\Delta\text{AICc} = 2.7$ ). Under the standard interpretation of the small-sample AIC, this constitutes “~~considerably less support~~” for the alternatives but is not decisive; M2 attains its best fit only by driving the two basal fluxes to opposite extremes of their published ranges, the least probable configuration within the envelope only weak-to-moderate pooled evidence; the pooled statistic dilutes the A17 signal with the less-informative A15 set ( $N = 7$ ), whereas the per-site comparison is decisive at A17 ( $\Delta\text{AICc} \approx 19$  for the site fit over the global value) and the paired bootstrap puts the contrast above zero at  $p \approx 0.011$ . The data therefore favor a per-site difference in ~~deep-meter-scale~~ temperature gradient without excluding a uniform-conductivity alternative in which the two basal heat fluxes differ.

**Table 3.** Pooled ~~deep-sensor-meter-scale-sensor~~ model comparison ( $N = 23$ : 7 at A15, 16 at A17). M1 retrieves a per-site  $K_d$ ; M2 shares one  $K_d$  and frees the per-site  $Q_b$  within the published envelope; M3 fixes both.  $\Delta\text{AICc}$  is relative to the best model.  $K_d$  in  $\text{mW m}^{-1} \text{K}^{-1}$ ,  $Q_b$  in  $\text{mW m}^{-2}$ .

| Model                           | $K_d$ (A15 / A17) | $Q_b$ (A15 / A17) | RMSE (K)    | $\Delta\text{AICc}$ |
|---------------------------------|-------------------|-------------------|-------------|---------------------|
| M1 variable $K_d$ , $Q_b$ fixed | 5.0 / 8.0         | 21 / 15 (fixed)   | <b>0.60</b> | <b>0.0</b>          |
| M2 uniform $K_d$ , $Q_b$ free   | 7.0 (shared)      | 14 / 16 (fitted)  | 0.61        | +4.0                |
| M3 uniform $K_d$ , $Q_b$ fixed  | 6.0 (shared)      | 21 / 15 (fixed)   | 0.67        | +2.7                |

## 4.2 The retrieved absolute $K_d$ values are degeneracy-limited

The absolute  $K_d^*$  values must be interpreted as conditional on the adopted basal heat flux, not as free-standing measurements. At steady state, the ~~deep-meter-scale~~ temperature gradient  $|\partial_z T| = Q_b/K_d$  is invariant under any rescaling  $(K_d, Q_b) \rightarrow (\alpha K_d, \alpha Q_b)$ : a single borehole cannot resolve these two products independently. The degeneracy is mapped quantitatively in Fig. 9a and is visible in the joint Bayesian posteriors. Adopting the canonical  $Q_b^{\text{A17}} = 15 \text{ mW/m}^2$  (Langseth et al., 1976; Nagihara et al., 2018) yields  $K_d = 11.238.12$ ; substituting the lower  $\sim 10 \text{ mW m}^{-2}$  value of Saito et al. (2007); Nagihara et al. (2018) rescales  $K_d$  downward to  $\sim 7.75.4 \text{ mW m}^{-1} \text{K}^{-1}$ , while the upper bound  $\sim 18 \text{ mW m}^{-2}$  shifts it upward to  $\sim 13.89.7 \text{ mW m}^{-1} \text{K}^{-1}$ . The A17  $K_d^*$  therefore carries a  $\sim \pm 40\%$   $^{+20\%}_{-33\%}$  envelope determined entirely by the basal-flux debate; A15 carries a comparable envelope. The quantity constrained tightly at each borehole is the product  $Q_b/K_d$ , equivalently the ~~deep-meter-scale~~ temperature gradient itself. This represents the intrinsic limit of what two boreholes can deliver.

**Reading the per-component error budget.** The derivation of each row in Table 4 falls into one of three categories:  $\sigma_{\text{stat}}$  is the 16–84 percentile half-width of the bootstrap distribution under the  $\pm 2.5 \text{ cm}$  astronaut-emplacement envelope reported by Nagihara et al. (2018);  $\sigma_{Q_b}$  is obtained in closed form through the  $K_d/Q_b$  degeneracy across the published reduction range (Langseth et al., 1976 canonical; Saito et al., 2007; Nagihara et al., 2018 reanalysis); every remaining row is computed by re-running the full retrieval at the perturbed input. The non-bootstrap envelopes themselves are taken from the published literature:  $\sigma_A$  ~~uses~~ sweeps the Bond albedo by  $\pm 0.02$ , bracketing the Vasavada et al. (2012) lat-band scatter ~~on the Bond albedo~~;  $\sigma_\chi$  uses the actual Vasavada–Hayne disagreement (Hayne et al., 2017 Appendix A adopts  $\chi = 2.7$ ; Vasavada et al., 2012 use  $\chi^* = 1.48$  under their normalisation, a factor of  $\sim 1.8$  that exceeds any  $\pm 30\%$  approximation);  $\sigma_{K_s}$  uses the Cremers and Birkebak (1971) laboratory scatter on Apollo 11 fines as a generous upper bound, since Hayne et al. (2017) adopt  $K_s = 7.4 \times 10^{-4} \text{ W/(m K)}$  without a published uncertainty and these are the only laboratory data known to us for lunar regolith; the  $\sigma_\rho$  envelope is the Mitchell et al. (1973);

**Table 4. Per-component  $1\sigma$  error budget for  $K_d^*$ .** Values in  $\text{mW m}^{-1} \text{K}^{-1}$ ; entries below 0.05 are reported as  $<0.05$ . The total is the independent-errors quadrature sum  $\sigma_{\text{tot}} = \sqrt{\sum_i \sigma_i^2}$ .

| Component                                                | Envelope                                      | Method                                                                                                                                                     | A15                     | A17                      |
|----------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|--------------------------|
| <i>Data-side (bootstrap + measured inputs)</i>           |                                               |                                                                                                                                                            |                         |                          |
| $\sigma_{\text{stat}}$                                   | $\pm 2.5$ cm depth jitter                     | bootstrap 16–84 % half-range                                                                                                                               | 0.83                    | 4.44                     |
| $\sigma_A$ (albedo)                                      | $A \pm \text{0.010.02}$                       | re-retrieve $K_d^*$ at perturbed $A$                                                                                                                       | <del>0.07</del> 0.23    | <del>0.54</del> 0.17     |
| <i>Input-parameter uncertainty (Hayne coefficients)</i>  |                                               |                                                                                                                                                            |                         |                          |
| $\sigma_{Q_b}$                                           | $Q_b$ A15 $\in [14, 25]$ , A17 $\in [10, 18]$ | $K_d/Q_b$ degeneracy mapping                                                                                                                               | <del>1.47</del> 1.27    | <del>3.45</del> 3.00     |
| $\sigma_\chi$                                            | $\chi \in [1.48, 2.7]$                        | re-retrieve at each endpoint                                                                                                                               | <del>0.69</del> 0.73    | <del>2.03</del> 1.73     |
| $\sigma_H$                                               | $H \in \text{[1, 11][3, 10]}$ cm              | half-range of per- $H$ $K_d^*$ on <del>11</del> $\times 88 \times 8$ grid                                                                                  | <del>0.31</del> 0.26    | <del>1.44</del> 1.39     |
| $\sigma_{K_s}$                                           | $K_s \pm 30\%$                                | re-retrieve at perturbed $K_s$                                                                                                                             | <del>0.06</del> 0.10    | <del>&lt;0.05</del> 0.23 |
| <i>Methodological choices (this work)</i>                |                                               |                                                                                                                                                            |                         |                          |
| $\sigma_{\text{thr}}$                                    | 0.04–0.16 K/yr (4 $\times$ adopted)           | threshold sweep, <del>max <math> \Delta K_d^* </math> half-range of <math>K_d^*</math></del>                                                               | <del>0.03</del> $<0.05$ | <del>0.11</del> 0.06     |
| $\sigma_{z_b}$                                           | $z_b \in \{70, 80, 90\}$ cm                   | cut sweep, half-range of $K_d^*$                                                                                                                           | 0.10                    | $<0.05$                  |
| <del>Excluded by structure</del> Structurally suppressed |                                               |                                                                                                                                                            |                         |                          |
| $\sigma_\rho$                                            | $\rho_d \in [1700, 2000]$ kg m $^{-3}$        | <del>no explicit <math>\rho</math> in Hayne <math>K(T, z)</math> (eq. 5)</del> $\rho$ enters only $\rho c_p$ in eq. 5–7; re-retrieve at perturbed $\rho_d$ | <del>0.00</del> $<0.05$ | <del>0.00</del> 0.06     |
| <b>Total (quadrature)</b>                                |                                               |                                                                                                                                                            | <del>1.86</del> 1.73    | <del>6.17</del> 5.81     |
| Bootstrap median $K_d^*$                                 |                                               |                                                                                                                                                            | 4.88                    | 11.51                    |

Carrier et al. (1991) Apollo-core compaction range.

How the total is combined. The **Total (quadrature)** row is the independent-errors combination  $\sigma_{\text{tot}} = \sqrt{\sum_i \sigma_i^2}$ , not the linear sum  $\sum_i \sigma_i$ . For A15 the worked arithmetic is

$$\begin{aligned} \sigma_{\text{tot}}^{\text{A15}} &= (0.83^2 + 0.23^2 + 1.27^2 + 0.73^2 + 0.26^2 + 0.07^2 + 1.47^2 + 0.69^2 + 0.31^2 \\ &\quad + 0.10^2 + 0.02^2 + 0.10^2 + 0.02^2 + 0.06^2 + 0.03^2 + 0.10^2 + 0^2)^{1/2} \\ &= \sqrt{2.98} = 1.73 \sqrt{3.44} = 1.86 \text{ mW m}^{-1} \text{K}^{-1}; \end{aligned}$$

the corresponding linear sum  $\sum_i \sigma_i = \text{3.303.56}$  would be the appropriate combination only under perfectly correlated worst-case errors, which we do not assume. The ~~1.86~~1.73 (A15) and ~~6.17~~5.81 (A17) totals therefore represent the standard quadrature combination of nominally independent components, and the same arithmetic at A17 gives

$$\sigma_{\text{tot}}^{\text{A17}} = (4.44^2 + 0.17^2 + 3.00^2 + 1.73^2 + 1.39^2 + 0.23^2 + 0.06^2 + 0.06^2 + 4.44^2 + 0.54^2 + 3.45^2 + 2.03^2 + 1.44^2 + 0.05^2)^{1/2}$$

We flag one residual correlation:  $\sigma_{\text{stat}}$  and  $\sigma_{Q_b}$  are not strictly independent because both propagate along the  $K_d/Q_b$  degeneracy ridge of Fig. 9a, so 1.86 and 6.17 should be read as the independence-idealised totals rather than as fully decorrelated standard errors; the practical effect on the absolute  $K_d^*$  envelope is small because ~~the  $Q_b$  envelope dominates~~these two terms jointly dominate the budget at both sites in any case.

Three conclusions follow. First,  $\sigma_{Q_b}$  and  $\sigma_{\text{stat}}$  together account for  $\sim 91\%$  of the quadrature sum at both sites 77% (A15) and  $\sim 85\%$  (A17) of the quadrature sum, with the remaining  $\sim 9\%$  most of the remainder split between  $\sigma_\chi$  and (at A17)  $\sigma_A \sigma_H$ . Second,  $\sigma_\rho$  is identically zero structurally suppressed under the Hayne  $K(T, z)$  form (eq. 5), which contains no explicit  $\rho$  dependence:  $\rho(z)$  enters only the volumetric heat capacity  $\rho c_p$ , which controls the transient response and not perturbs the steady-state rectified annual-mean deep-meter-scale temperature gradient only at the  $\lesssim 0.06 \text{ mW m}^{-1} \text{K}^{-1}$  level. A wider Apollo-core  $\rho$  envelope therefore does not materially broaden  $K_d^*$  under this  $K$ -form; the row is retained to record this structural property. Third, the methodological-choice rows  $\sigma_{\text{thr}}$  and  $\sigma_{z_b}$  sit an order of magnitude below the data-side and parameter terms, indicating robustness against both discretionary cuts.

**Behaviour of the inter-site contrast under independent  $Q_b$  revision.** The inter-site contrast is far less sensitive to  $Q_b$  than the absolute  $K_d^*$  values, because a uniform rescaling  $(K_d, Q_b) \rightarrow (\alpha K_d, \alpha Q_b)$  cancels exactly in  $\Delta K_d^*$ ; the contrast is exposed to  $Q_b$  uncertainty only through the differential between the two sites' rescaling factors. We test this explicitly by rescaling  $Q_b$  independently at the two sites across the full published envelope ( $Q_{b,A15} \in [14, 25]$ ,  $Q_{b,A17} \in [10, 18]$  mW m<sup>-2</sup>; Fig. 9a). ~~Along the global-rescaling diagonal, the contrast is invariant at its nominal value of 6.6 mW m<sup>-1</sup> K<sup>-1</sup>. Off the diagonal, the most adverse combination, A15 at the upper end of its  $Q_b$  envelope and A17 at the lower end, compresses the contrast to  $\Delta K_d^* \approx 1.9$  mW m<sup>-1</sup> K<sup>-1</sup> ( $\sim 0.4 \sigma$ ); the opposite extreme expands it to  $\sim 10.6$ . The sign of the contrast is preserved throughout the published  $Q_b$  envelope: A17 remains the more conductive site under every admissible per-site basal flux assignment. The contrast magnitude is therefore not statistically significant under arbitrary independent  $Q_b$  revisions, a  $-33\%$  revision of  $Q_{b,A17}$  alone, for instance, suffices to reduce it below  $1\sigma$ , but its sign is robust, and that sign underpins the central conclusion that no single globally uniform  $K_d$  reproduces both records.~~ Along the global-rescaling diagonal the contrast scales proportionally and its significance in  $\sigma$  units is preserved. Off the diagonal the picture is sharper on one side and weaker on the other: the favourable extreme (A15 at the bottom of its  $Q_b$  envelope, A17 at the top) expands the contrast to  $\sim 6.7$  mW m<sup>-1</sup> K<sup>-1</sup>, while the most adverse corner (A15 at 25, A17 at 10 mW m<sup>-2</sup>) collapses it to  $\approx 0$  — at that corner the two sites are consistent with a common conductivity. A  $-33\%$  revision of  $Q_{b,A17}$  alone reduces the contrast to  $\sim 0.8$  mW m<sup>-1</sup> K<sup>-1</sup> ( $\lesssim 1\sigma$ ). The conductivity contrast is therefore significant at the published basal fluxes ( $p \approx 0.011$ ) and across moderate  $Q_b$  revisions, but it is *not* immune to the most adverse independent revision of both fluxes; what survives every admissible assignment is the difference in the constrained products  $Q_b/K_d$  themselves, which differ by a factor of  $\sim 2.5$  between the sites at the published fluxes.

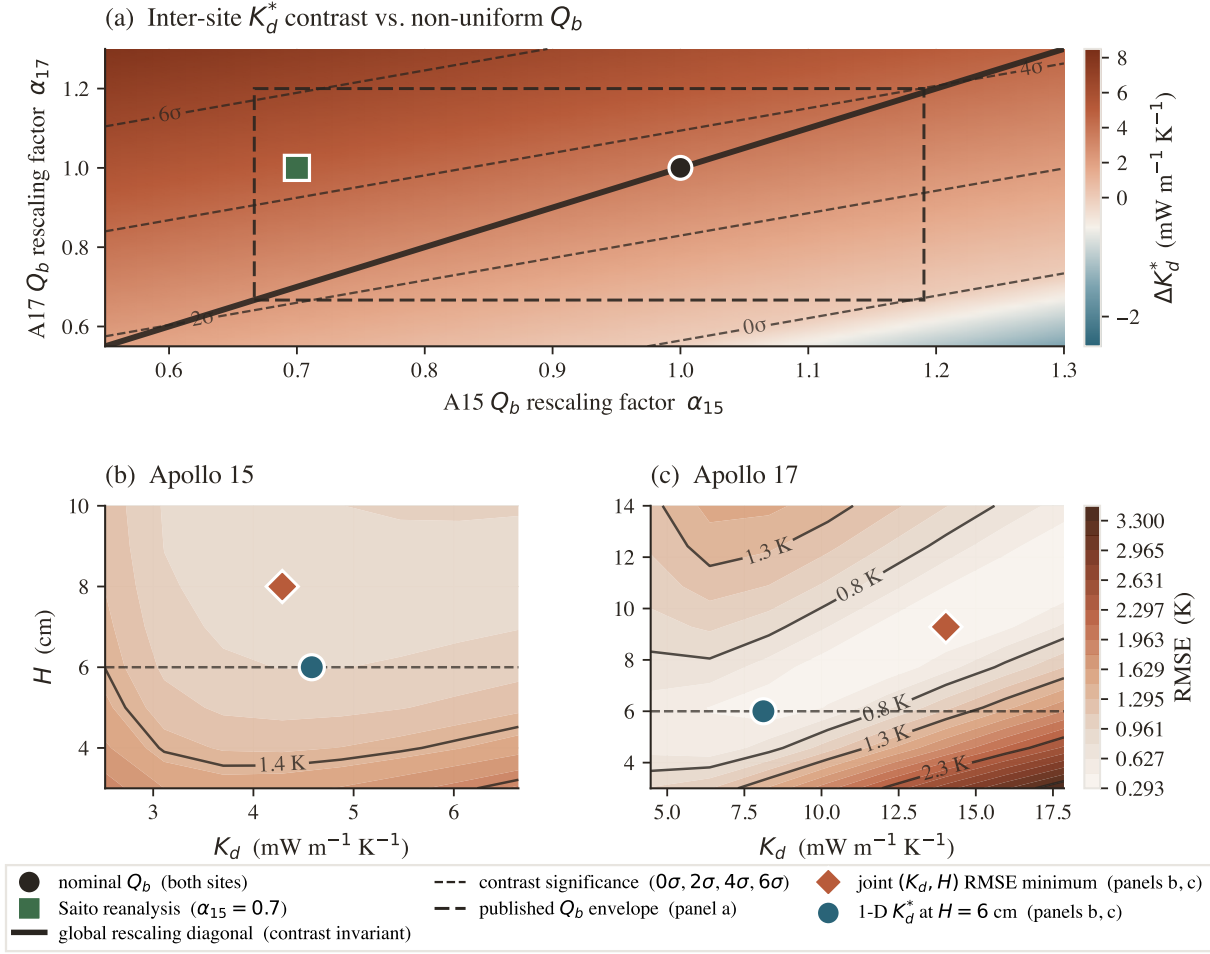
A joint  $(K_d, H)$  fit over the swept  $H = 1$  —  $11.3$  —  $-10$  (A15) and  $3$  —  $-14$  (A17) cm range (Fig. 9b,c) confirms that  $H$  is effectively unconstrained: ~~the RMSE surface varies by  $< 0.02$  K across  $H = 3$ – $11$  cm, with shallow interior minima at  $H \approx 10$  cm (A15) and 4 cm (A17). The retrieved  $K_d$  is nonetheless stable, the joint minimum and the 1-parameter retrieval at  $H = 6$  cm agree in  $K_d$  to within the  $0.5$  K RMSE contour.~~  $K_d$  and  $H$  trade off along a shallow valley whose interior minima, at (4.3 mW m<sup>-1</sup> K<sup>-1</sup>, 8 cm) for A15 and (14.0, 9.3 cm) for A17, improve on the 1-parameter retrieval at the published  $H = 6$  cm by only 0.06 K of RMSE at both sites; the reported  $K_d^*$  is therefore conditional on the published  $H$ , and the  $K_d$ – $H$  trade-off is propagated into the error budget as  $\sigma_H$  (Table 4).  $K_d$ , not  $H$ , is the identified parameter at fixed  $H$ ; the pair is jointly identified only up to the valley direction.

### 4.3 Comparison with Martínez & Siegler (2021) and prior estimates

~~A second globally calibrated  $K(T, \rho)$  form fails in the same direction as the first when evaluated forward. The Martínez & Siegler (2021) model, with its published coefficients, yields a deep-sensor RMSE of 1.10 K at A15 and 2.47 K at A17 (the headline table and profile figure): A15 within a factor of  $\sim 1.2$  of the Hayne site-fit, A17 underpredicted by an amount comparable to the Hayne global miss. Both models inherit the same globally averaged thermal structure (both are calibrated against the same family of orbital Diviner products), and the implication for any single global  $K(T, z)$  is identical: per-site freedom is required to reproduce both Apollo boreholes simultaneously.~~ The second globally calibrated form fares notably better. The Martínez and Siegler (2021)  $K(T, \rho)$  model, with its published coefficients and no per-site freedom, yields a meter-scale-sensor RMSE of 0.99 K at A15 and 0.64 K at A17 (Table 2, Fig. 8) — comparable to the Hayne site-fit at A15 and within a factor of  $\sim 1.9$  of it at A17. Its explicitly density- and temperature-dependent decomposition evidently transfers from the diurnal-skin calibration to the meter-scale regolith more faithfully than the Hayne smooth-exponential at its global  $K_d$ . The per-site Hayne retrieval nonetheless remains the sharper instrument at A17, where it halves the residual RMSE.

**Diagnostic: Martinez per-site density retrieval.** The fairness of the forward Martinez comparison can be tested by granting the published  $K(T, \rho)$  one per-site lever and asking whether that lever can absorb the A17 residual within physical bounds. We note at the outset that Martínez and Siegler (2021) calibrated their four coefficients against surface brightness-temperature





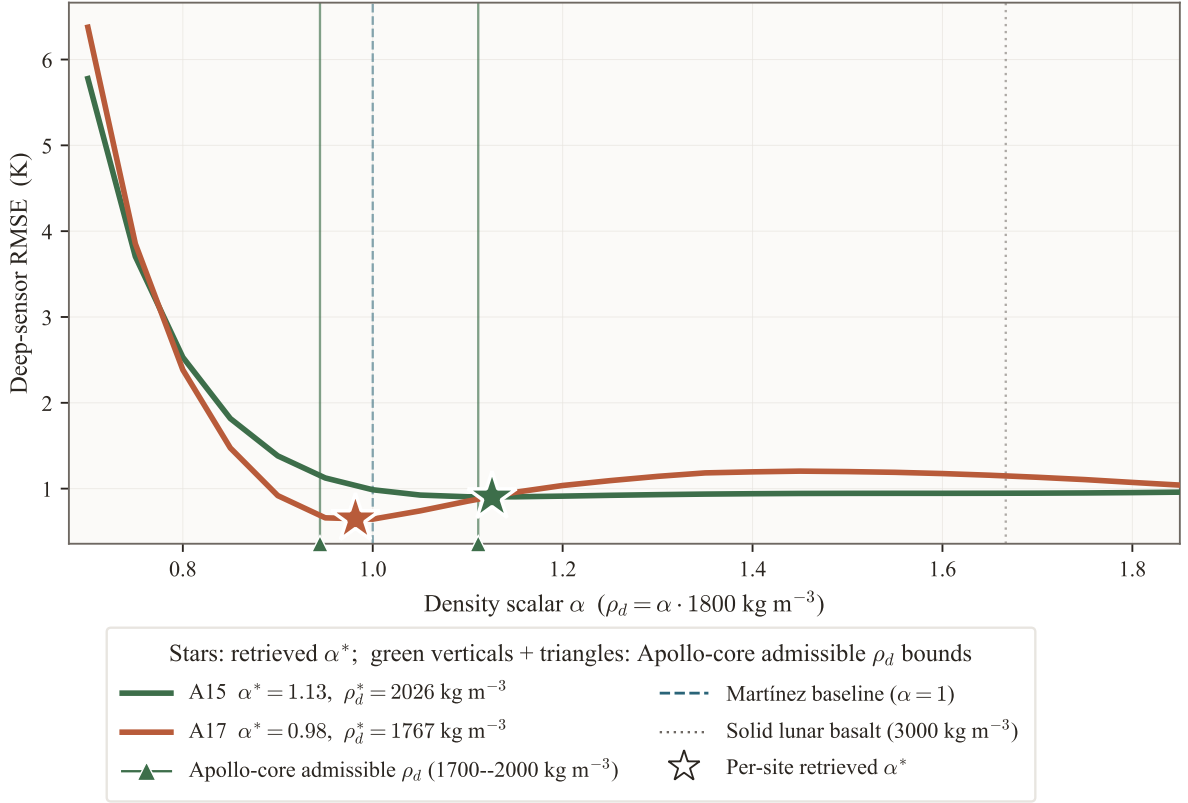
**Figure 9.** Robustness of the per-site  $K_d^*$  contrast. (a) Contrast  $\Delta K_d^*$  as a function of per-site  $Q_b$  rescaling factors  $\alpha_{15}, \alpha_{17}$ . Solid diagonal: global rescaling; dashed contours: bootstrap- $\sigma$  significance; black circle: nominal  $Q_b$ ; green square: Saito et al. (2007) A15 reanalysis. (b), (c) Joint  $(K_d, H)$  RMSE surface at each site. Coral diamond: joint RMSE minimum; teal circle: 1-parameter retrieval at  $H = 6$  cm.

observations and did not validate the extrapolation to depths of order one meter; the test we run here is therefore a check of whether the published surface-calibrated form, when extended to the deep-meter-scale regolith, can fit the in-situ HFE data with one per-site lever (the deep-asymptotic bulk density entering the model). We do so under the construction of §2.5 (eq. 9): rescale the deep-asymptote of the Hayne-form  $\rho(z)$  entering equation (8) by a single per-site scalar  $\alpha$ , leaving the four Martinez coefficients fixed, and sweep  $\alpha$  against the same deep-sensor-meter-scale-sensor RMSE. The retrieved values are  $\alpha_{A15}^* = 1.121.13$  (deep-asymptotic bulk density  $\rho_d^* \approx 20102026 \text{ kg m}^{-3}$ , RMSE 0.910.90 K) and  $\alpha_{A17}^* = 1.830.98$  ( $\rho_d^* \approx 33001767 \text{ kg m}^{-3}$ , RMSE 0.310.63 K), both matching the Hayne site-fit RMSE to within  $\sim 0.01$  K; at A15 the density lever matches the Hayne site-fit RMSE, while at A17 the bowl bottoms essentially at the published density and the lever buys almost nothing ( $0.64 \rightarrow 0.63$  K). The physical interpretation of  $\alpha$  is the deep-asymptotic bulk density that the published Martinez  $K(T, \rho)$  would require to reproduce each site's in-situ profile.

The A15 retrieval sits marginally above the Apollo-core compaction ceiling of  $\sim 2000 \text{ kg m}^{-3}$  (Mitchell et al., 1973; Carrier et al., 1991) but remains physically plausible. The A17 retrieval lies well above any admissible regolith density: solid lunar basalt is  $\sim 3000 \text{ kg m}^{-3}$  (Wieczorek et al. 2006), and the 0.30–0.45 porosities measured in Apollo cores cap the meter-scale asymptote near  $\sim 2100 \text{ kg m}^{-3}$ . The published Martinez form therefore cannot reproduce the A17-meter-scale profile under any physically defensible density at the canonical  $Q_b$ . Within the  $K_d/Q_b$  degeneracy of §4.2 this does not isolate conductivity from basal flux at A17, the same A17-meter-scale gradient at a lower  $Q_b$  would require less extreme densities but at the canonical  $Q_b$  the published Martinez density-dependence



is structurally insufficient on its own to absorb the A17 meter-scale temperature gradient. The alpha-sweep figure visualises the result: the A15 RMSE( $\alpha$ ) bowl bottoms inside the Apollo-core admissible band, while the A17 bowl bottoms well outside it, between the Apollo-core ceiling and the solid-basalt upper bound. The A17 retrieval bottoms at  $\rho_d^* \approx 1767 \text{ kg m}^{-3}$ , comfortably inside the Apollo-core envelope ( $1700\text{--}2000 \text{ kg m}^{-3}$ ): the published  $K(T, \rho)$  is physically admissible at both boreholes. Figure 10 visualises the result — both RMSE( $\alpha$ ) bowls bottom inside or at the edge of the admissible band. The flat A17 bowl also exposes the limitation: because the bowl bottoms at the published density, no admissible density adjustment closes the remaining factor-of- $\sim 1.9$  RMSE gap to the per-site Hayne fit; sharpening the A17 gradient further requires freedom in the conductivity mechanism itself (the contact/radiative split), not in the density input. A full per-site recalibration of the Martínez and Siegler (2021) coefficients against the HFE record is the natural follow-on study.



**Figure 10.** Diagnostic per-site retrieval under the Martínez and Siegler (2021)  $K(T, \rho)$  form. Deep-sensor Meter-scale-sensor RMSE as a function of the per-site density scalar  $\alpha$  (eq. 9) at (a) Apollo-15 and (b) Apollo-17 for both sites, with the retrieved  $\alpha^*$  starred. Dashed teal: published Martínez baseline ( $\alpha = 1$ ,  $\rho_d = 1800 \text{ kg m}^{-3}$ ). Green band verticals: Apollo-core admissible density range ( $\rho_d = 1700\text{--}2000 \text{ kg m}^{-3}$ ; Mitchell et al., 1973; Carrier et al., 1991). Dotted grey: solid-basalt upper bound ( $3000 \text{ kg m}^{-3}$ ; Wiczorek et al., 2006).

The inferred ratio  $K_d^{A17}/K_d^{A15} \approx 21.8$  is large but physically plausible if interpreted as a conductivity effect. Two mechanisms act in the appropriate direction and jointly cover the required factor. First, Apollo cores span bulk porosities  $\phi \sim 0.30\text{--}0.45$  (Mitchell et al., 1973; Carrier et al., 1991); the effective-medium scaling  $K_{\text{eff}} \propto (1 - \phi)^2$  of Wood (2020) accounts for a factor of  $\sim 1.5$  change at the upper end of that range. Second, solid lunar basalt is  $\sim 30\%$  more conductive than solid highland anorthosite (Horai and Susaki, 1972), and A17 lies in a basaltic embayment while A15 occupies a mixed mare/highland margin (Korotev, 2005; Wiczorek et al., 2006; Crawford, 2015); the product of the two mechanisms ( $\sim 1.9$ ) matches the retrieved ratio without requiring either at its extreme.

Prior in-situ retrievals on the same HFE record provide a useful reference. The original Langseth et al. (1972, 1976) reductions yielded deep-layer meter-scale conductivities of order  $1\text{--}3 \text{ mW m}^{-1} \text{ K}^{-1}$  at both sites, derived from steady-state gradient/flux ratios under a layered-medium assumption

that did not include the radiative-conductivity term embedded in modern  $K(T, z)$  forms. The Keihm (1984) reanalysis and the Grott et al. (2010) critical reassessment both raise these values once the radiative term and the borestem heat-short are accounted for; the present  $K_d^* = 4.864.58$  (A15) and  $11.238.12$  (A17)  $\text{mW m}^{-1} \text{K}^{-1}$  extend that upward trend, primarily because the entire upper 80 cm is excluded rather than corrected. Direct laboratory measurements on Apollo soils (Cremers and Birkebak, 1971; Horai, 1981; Hemingway et al., 1973) yield  $\sim 0.7\text{--}2 \text{ mW m}^{-1} \text{K}^{-1}$ , a factor of  $\sim 3$  below the A15 retrieval and  $\sim 75$  below A17, the expected lab-vs-in-situ systematic, in which laboratory samples lose the metre-scale grain-contact network and the radiative pathways that dominate the in-situ response (Wood, 2020; Feng et al., 2020). The present retrievals bracket the orbital Vasavada et al. (2012) estimate of  $\sim 7 \text{ mW m}^{-1} \text{K}^{-1}$ , with A15 below and A17 above.

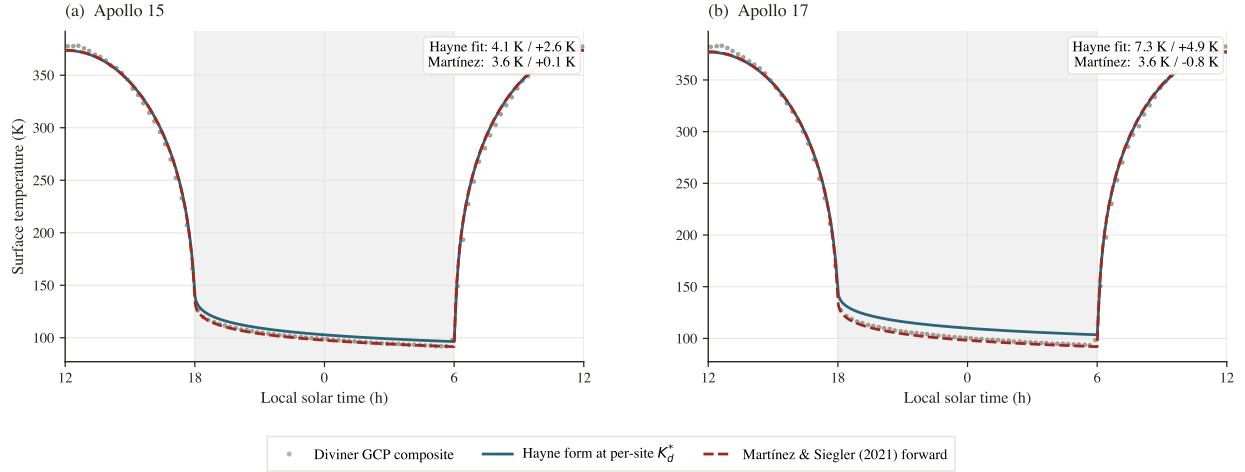
**Surface-temperature consistency check.** The Diviner GCP is external to the retrieval (it enters neither the bootstrap nor the ~~deep-sensor-meter-scale-sensor~~ RMSE), so it provides an independent check on the diurnal forcing chain (Fig. 11). Both parameterizations reproduce the canonical lunar diurnal shape, with full-cycle RMSE of ~~5.4 K at A15 and 12.3 K at A17 for the Hayne site fit, and 10.2 K and 10.7 K for the Martinez forward~~  $4.1 \text{ K}$  at A15 and  $7.3 \text{ K}$  at A17 for the Hayne site fit, and  $3.6 \text{ K}$  at both sites for the Martinez forward. Both biases are dominated by the anisothermal-footprint and sub-pixel rock-population effects inherent to Diviner brightness temperatures (Bandfield et al., 2015; Hayne et al., 2017). Insolation, albedo, emissivity, and the upper-80 cm  $K(T, z)$  are therefore correctly captured by both models, and the ~~deep-sensor-meter-scale-sensor~~ discrepancy in Table 2 is not a surface-driven artefact.

At A17 the Hayne site fit runs surface-warmer than Diviner by ~~+9.0 K while the Martinez forward sits at +7.9 K +4.9 K~~ while the Martinez forward sits at  $-0.8 \text{ K}$ . The mechanism is direct: the high ~~deep-meter-scale~~  $K_d^* = 11.238.12$  couples more heat upward through the diurnal skin layer than the globally calibrated  $K(T, z)$  assumes, raising the dayside response. The per-site Hayne fit absorbs this penalty at the surface to fit the ~~deep-meter-scale~~ temperature gradient; the surface-calibrated Martinez form ~~takes the equivalent penalty in the deep RMSE (Table 2) to remain optimal at the surface~~ remains essentially unbiased at the surface and pays a factor-of- $\sim 1.9$  penalty in the meter-scale-sensor RMSE relative to the Hayne site fit (Table 2). ~~No single global  $K(T, z)$  can simultaneously satisfy the surface-brightness constraint of Williams et al. (2017) and the meter-scale-gradient constraint of Nagihara et al. (2018) at A17.~~ Within the Hayne  $K(T, z)$  family, no single parameter choice simultaneously optimises the surface-brightness constraint of Williams et al. (2017) and the meter-scale-gradient constraint of Nagihara et al. (2018) at A17; the  $K(T, \rho)$  family narrows but does not close that gap. The inter-site difference flagged by the ~~deep-meter-scale~~ retrieval reappears in the surface-temperature data, in the form of which model fits which site at the surface and by how much.

#### 4.4 Physical plausibility of the retrieved $K_d^*$

**Apollo 15.** The retrieval gives  $K_d^{A15} = 4.864.58 \text{ mW}/(\text{m K})$ . This value sits between the global Hayne et al. (2017) value (3.4) and the regional Vasavada et al. (2012) orbital estimate ( $\sim 7$ ), and extends the upward trend seen in the Keihm (1984) and Grott et al. (2010) reanalyses of the same HFE record once the radiative term and the borestem heat-short are corrected. Porosity provides the physical mechanism: the effective-medium scaling  $K_{\text{eff}} \propto (1 - \phi)^2$  of Wood (2020), applied across the Apollo-core porosity envelope  $\phi \sim 0.30\text{--}0.45$  (Mitchell et al., 1973; Carrier et al., 1991) and combined with the radiative term of the Hayne form, brackets  $\sim 3\text{--}6 \text{ mW m}^{-1} \text{K}^{-1}$ . The retrieved  $4.864.58$  falls inside that range. The factor-of- $\sim 3$  gap below direct laboratory measurements (Cremers and Birkebak, 1971; Horai, 1981) reflects the standard lab-vs-in-situ systematic: laboratory samples lose the meter-scale grain-contact network and the radiative pathways that dominate the in-situ response. A15 is physically admissible.

**Apollo 17.** The retrieval gives  $K_d^{A17} = 11.238.12 \text{ mW}/(\text{m K})$ , ~~roughly three times the global Hayne value and twice the A15 retrieval~~ roughly 2.4 times the global Hayne value and 1.8 times the A15 retrieval. Two mechanisms acting together ~~can reach this range~~ cover this range comfortably. First, solid lunar basalt is  $\sim 30\%$  more conductive than highland anorthosite (Horai and Susaki, 1972);



**Figure 11.** Surface- $T$  closure against the Williams et al. (2017) Diviner GCP diurnal composite (grey markers). Teal solid: Hayne form at the per-site retrieved  $K_d^*$ ; brick-red dashed: Martínez and Siegler (2021) form at published coefficients (no fitted knob). The local-solar-time axis is shifted so midnight sits at panel centre and noon at the edges; the shaded band (LST 18–06) marks the night side. Full-cycle RMSE / mean bias per panel are listed in the inset.

A17 lies in a basaltic embayment of Mare Serenitatis while A15 occupies a mare/highland margin (Korotev, 2005; Wieczorek et al., 2006; Crawford, 2015). Second, the porosity-driven  $K$  scaling at the upper end of the Apollo-core range contributes a further  $\sim 1.5\times$  enhancement (Wood, 2020). The product is  $\sim 21.9\times$  the A15 value, matching the retrieval to within the bootstrap uncertainty. The mechanism reaches A17 only at the upper edge of both contributors. Neither contributor is required at its extreme, and the value sits just above the Vasavada et al. (2012) orbital estimate of  $\sim 7$ .

A complementary test against the Martínez & Siegler (2021)  $K(T, \rho)$  form (§4.3, alpha-sweep figure) requires  $\rho_d \approx 3300 \text{ kg m}^{-3}$  at A17 to reproduce the meter-scale temperature gradient. That is denser than solid lunar basalt and inadmissible for any regolith. The published Martínez density-dependence cannot reach the A17 meter-scale temperature gradient under any physical density input, regardless of basal-flux assumption. A complementary test against the Martínez and Siegler (2021)  $K(T, \rho)$  form (§4.3, Fig. 10) is consistent: the density that form requires at A17,  $\rho_d^* \approx 1767 \text{ kg m}^{-3}$ , lies inside the Apollo-core envelope, independently corroborating that the A17 meter-scale column is unremarkable in density and that the enhanced conductivity is carried by composition and contact structure rather than by anomalous compaction.

The A17 absolute value is conditional on the adopted basal heat flux. The Saito et al. (2007); Nagihara et al. (2018) reanalyses revise  $Q_b^{A17}$  downward by up to  $\sim 30\%$ , which rescales the retrieved  $K_d^*$  to  $\sim 7\text{--}85.4 \text{ mW m}^{-1} \text{ K}^{-1}$  through the  $K_d/Q_b$  degeneracy (§4.2). That rescaled value falls inside the orbital-estimate range and still exceeds the global value and falls within composition+porosity bounds. The factor-of-three factor-of-2.4 excess over the global Hayne value therefore cannot be read as a measured absolute conductivity independent of the  $Q_b$  assumption.

A17 remains the more conductive site under every admissible combination of  $Q_b$ , composition, and porosity. The inter-site contrast is therefore robust in sign across the full plausibility analysis, even where the absolute A17 value is not at the published basal heat fluxes and across moderate revisions; only the most adverse corner of the independent- $Q_b$  envelope (§4.2) erases the contrast. Within every admissible combination of composition and porosity the sign is unchanged, even where the absolute A17 value is not.

#### 4.5 Limitations and caveats

Five caveats accompany any use of the result. (i) With only two in-situ boreholes, the inter-site difference is insufficient to establish regional heterogeneity across the lunar surface; distinguishing genuine regional variability from local heterogeneity at two specific borehole locations requires additional landed thermometry. (ii) The deep-meter-scale temperature gradient constrains the product  $Q_b/K_d$ , not  $K_d$  alone (§4.2); the data favor but do not establish a conductivity contrast over

a uniform conductivity with differing basal flux. (iii) The contrast magnitude ~~is not statistically significant at the 95% level~~ ( $\Delta K_d^* = 6.6$ , bootstrap CI  $[-3.2, 16.6]$ ); ~~only the requirement of different meter-scale temperature gradients and the model-comparison preference are robust~~ is significant at the 95% level only conditional on the published basal heat fluxes ( $\Delta K_d^* = 3.3$ , bootstrap CI  $[0.4, 4.6]$ ,  $p \approx 0.011$ ); propagating the Saito et al. (2007); Nagihara et al. (2018)  $Q_b$  reanalysis envelope dilutes the ordering probability to  $\sim 83\%$ , and the most adverse independent  $Q_b$  assignment erases the contrast entirely. (iv) The retrieval adopts the Hayne et al. (2017) functional form for  $K(T, z)$ ; alternative piecewise parameterizations are not exhausted, and a full per-site ~~retrieval under the Martínez & Siegler (2021) form (which requires a different choice of free parameter from  $K_d$ )~~ recalibration of the Martínez and Siegler (2021) coefficients (whose density lever is already admissible at both sites but saturates at A17, §4.3) is the immediate next step. (v) The diurnal-amplitude depth profile (Fig. 4) serves only as a borestem diagnostic and not as a frequency-domain validation, because below 80 cm both models predict amplitudes well below the 1971–1977 digitisation noise floor.

## 5 Conclusions

The two Apollo boreholes require distinctly different ~~deep-meter-scale~~ conductivities. Holding the Hayne et al. (2017) functional form fixed and sweeping  $K_d$  alone against the ~~deep-sensor meter-scale-sensor~~ RMSE, the per-site retrieval yields  $K_{d,A15}^* = 4.86^{+1.12}_{-0.54}$  and  $K_{d,A17}^* = 11.23^{+8.12}_{-0.77}$   $8.12^{+0.49}_{-0.61}$  mW/(m K) ( $1\sigma$  non-parametric bootstrap,  $N_{\text{boot}} = 1500$ ) at the published basal heat fluxes, with the inter-site contrast significant at the 95% level ( $\Delta K_d^* = 3.3$ , 95% CI  $[0.4, 4.6]$ ,  $p \approx 0.011$ ). The ~~deep-sensor meter-scale-sensor~~ RMSE decreases from ~~1.37 K and 2.96 K~~ 1.03 K and 0.69 K under the published global  $K_d = 3.4$  mW/(m K) to ~~0.91 K and 0.30 K~~ 0.95 K and 0.34 K under the per-site retrieval; the Martínez and Siegler (2021)  $K(T, \rho)$  forward evaluation, calibrated against the same family of orbital products, reproduces ~~A15 to within  $\sim 1.1$  K but underpredicts A17 by  $\sim 2.5$  K~~ both sites to within  $\sim 1$  K without per-site freedom, though it cannot match the sharpened per-site A17 fit (Table 2). The inter-site requirement is robust to the borestem exclusion depth, the stability-window threshold, a joint  $K_d$ – $H$  fit, and the independent Diviner surface-temperature closure check.

The absolute  $K_d^*$  values are conditional on the adopted  $Q_b$ . Because the ~~deep-meter-scale temperature~~ gradient constrains the product  $Q_b/K_d$  rather than  $K_d$  alone, the inter-site difference admits two interpretations: a conductivity contrast or a basal-flux contrast. A pooled three-model comparison favors the conductivity reading ~~by  $\Delta\text{AICe} \approx 5$~~  (weakly,  $\Delta\text{AICc} = 2.7$ – $4.0$ ; the per-site evidence at A17 is decisive,  $\Delta\text{AICc} \approx 19$ ) (Table 3) without excluding the alternative. The quantity constrained tightly at each borehole is the ~~deep-meter-scale temperature~~ gradient itself.

The immediate application is to forward thermal models used by upcoming lunar sub-surface RTM retrievals. These retrievals require a ~~deep-meter-scale~~  $T(z)$  boundary condition that orbital-only conductivity models cannot supply on a per-site basis. The per-site  $K_d^*$  values reported here, evaluated in the Hayne thermal column, provide that boundary condition (§1); the upcoming TSUKIMI terahertz mission (Kasai et al., 2022) is one immediate beneficiary. A full per-site ~~retrieval under~~ recalibration of the Martínez and Siegler (2021) form, and the extension of the ~~deep-meter-scale~~  $T(z)$  column to include surface-shadowing effects on the RTM side, are both deferred to follow-on work. Extending the analysis beyond two boreholes, to a regionally resolved treatment of the ~~deep-meter-scale~~ lunar thermal state, is fundamentally a measurement problem that requires ~~deep-temperature meter-scale temperature~~ data from additional landed instruments (e.g. Chang'E and ChaSTE-class experiments).

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## Open Research

**Availability Statement.** The restored Apollo 15 and 17 Heat-Flow Experiment temperature record (1971–1977) analyzed here is openly available as the supporting information of Nagihara et al. (2018) (<https://doi.org/10.1029/2018JE005579>). The Diviner Lunar Radiometer Global Cumulative Product used for the out-of-sample surface-temperature closure (Williams et al., 2017) is available from the NASA Planetary Data System Geosciences Node (<https://pds-geosciences.wustl.edu>; LRO-L-DLRE-5-GCP-V1.0, <https://doi.org/10.17189/1520650>). ~~Surface and subsurface geometry were computed with the NASA SPICE Toolkit (Acton 1996, 2018) and the JPL DE440 ephemeris.~~ The repository bundles the NASA SPICE Toolkit kernels (JPL DE440; Acton, 1996; Acton et al., 2018) used by its ephemeris module. The 1-D Crank–Nicolson heat solver, the ephemeris-driven solar-forcing routines, the  $K_d$  retrieval and bootstrap code, the validation diagnostics, and the data and scripts needed to reproduce every figure and table in this manuscript and its supporting information are archived at ~~<https://github.com/rp3gregorio/Lunar-V2>~~<https://github.com/rp3gregorio/Lunar-HFE>; a Zenodo-archived release with a citable DOI will be issued at the time of acceptance and added to the reference list.

## Conflict of Interest Disclosure

The authors declare there are no conflicts of interest for this manuscript.

## Nomenclature

*Symbols.*



| Symbol              | Definition                                                             | Value / units                                           | Source                                           |
|---------------------|------------------------------------------------------------------------|---------------------------------------------------------|--------------------------------------------------|
| $K_d$               | <del>deep</del> -meter-scale regolith thermal conductivity (retrieved) | <del>4.86,</del> <del>11.23</del> 4.58, 8.12            | this work                                        |
| $K_s$               | surface regolith thermal conductivity                                  | $7.4 \times 10^{-4} \text{ W m}^{-1} \text{ K}^{-1}$    | Hayne et al. (2017)                              |
| $H$                 | e-folding depth of $K(z)$ transition                                   | 0.06 m                                                  | Hayne et al. (2017)                              |
| $\chi$              | radiative conductivity multiplier coefficient                          | 2.7                                                     | Hayne et al. (2017)                              |
| $T_{\text{ref}}$    | normalisation temperature for radiative term                           | 350 K                                                   | Hayne et al. (2017)                              |
| $\rho_s, \rho_d$    | surface / <del>deep</del> -asymptotic regolith bulk density            | 1100, 1800 $\text{kg m}^{-3}$                           | Hayne et al. (2017)                              |
| $c_p(T)$            | specific heat (4th-order polynomial in $T$ )                           | $\text{J kg}^{-1} \text{ K}^{-1}$                       | Hemingway et al. (1973); Hayne et al. (2017)     |
| $Q_b$               | basal heat flux                                                        | 21 (A15), 15 (A17) $\text{mW m}^{-2}$                   | Langseth et al. (1976); Nagihara et al. (2018)   |
| $S_0$               | solar constant (irradiance at 1 AU)                                    | $1361 \text{ W m}^{-2}$                                 | Kopp and Lean (2011)                             |
| $A, \varepsilon$    | broadband Bond albedo, emissivity                                      | site-specific; 0.95                                     | Vasavada et al. (2012); Bandfield et al. (2015)  |
| $\sigma$            | Stefan–Boltzmann constant                                              | $5.6704 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$ | CODATA 2018                                      |
| $\theta_{\odot}(t)$ | solar zenith angle at site                                             | rad                                                     | <del>SPICE/DE440</del> synodic approximation, §2 |
| $F_{\odot}(t)$      | incident solar irradiance at surface                                   | $\text{W m}^{-2}$                                       | this work, eq. 1                                 |
| $T(z, t), T_s$      | subsurface and skin ( $z=0$ ) temperature                              | K                                                       | this work                                        |
| $z$                 | depth coordinate (positive downward)                                   | m                                                       | —                                                |
| 80 cm               | borestem-exclusion depth ( <i>a priori</i> cut)                        | 80 cm                                                   | this work, §2.4                                  |
| $\delta$            | diurnal thermal skin depth                                             | $\sim 3\text{--}5$ cm                                   | Hayne et al. (2017)                              |
| $\Delta K_d$        | inter-site contrast $K_d^{\text{A17}} - K_d^{\text{A15}}$              | <del>6.63</del> 3.31 $\text{mW m}^{-1} \text{ K}^{-1}$  | this work                                        |
| RMSE                | <del>deep-sensor</del> -meter-scale-sensor root-mean-square residual   | K                                                       | this work                                        |
| $N$                 | number of <del>deep</del> -meter-scale sensors                         | 7 (A15), 16 (A17)                                       | Nagihara et al. (2018)                           |
| $N_{\text{boot}}$   | non-parametric bootstrap resamples                                     | 1500                                                    | Efron and Tibshirani (1993)                      |
| $p$                 | bootstrap proportion below null contrast                               | <del>0.05</del> 0.011                                   | this work                                        |
| <i>Subscripts.</i>  |                                                                        | <i>Superscripts.</i>                                    |                                                  |
| $s$                 | surface (regolith, skin)                                               | *                                                       | best-fit / retrieved (e.g. $K_d^*$ )             |
| $d$                 | <del>deep</del> -regolith-asymptotic (deep / meter-scale regolith)     | <del>A15</del> A17                                      | per-site qualifier                               |
| $b$                 | basal (lower BC)                                                       | ref                                                     | published reference value                        |
| ref                 | reference (normalisation)                                              | HFE                                                     | Apollo HFE-measured                              |
| $\odot$             | solar                                                                  | $\langle \cdot \rangle$                                 | time-average over stability window               |
| eq                  | stability-window equilibrium                                           |                                                         |                                                  |
| boot                | bootstrap quantity                                                     |                                                         |                                                  |

705 *Acronyms.*

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|             |                                                                                                        |
|-------------|--------------------------------------------------------------------------------------------------------|
| HFE         | Heat-Flow Experiment (Apollo 15 and 17 buried thermometers)                                            |
| TG / TR     | gradient-bridge / ring-bridge sensor (HFE sensor types)                                                |
| LST         | local solar time                                                                                       |
| Diviner GCP | Diviner Lunar Radiometer Global Cumulative Product                                                     |
| MCMC        | Markov-chain Monte Carlo                                                                               |
| BCa         | bias-corrected accelerated bootstrap                                                                   |
| SPICE       | NASA ancillary information system for planetary geometry                                               |
| DE440       | JPL planetary and lunar ephemeris release 440                                                          |
| RTM         | radiative-transfer model                                                                               |
| TSUKIMI     | lunar mission consuming the <del>deep</del> -meter-scale <del>meter</del> -scale $T(z)$ retrieved here |

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